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Basic Electrical Quantities, Components, and Sources

{Overview

} This unit builds the basic language of electrical engineering, introducing the core quantities, passive components, waveform families, source models, and practical measurement ideas used throughout the book.

{Contents

}

Unit 1: Basic Electrical Quantities, Components, and Sources

1.1 Basic Electrical Quantities 1.2 Passive Components 1.3
Signal Waveforms 1.4 Sources

Unit 1: Basic Electrical Quantities, Components, and Sources

A torch, a ceiling fan, a mobile charger, and a water pump all raise the same underlying questions: what pushes charge through a circuit, what flows, why a component heats, and why a battery voltage falls when loaded. This chapter builds the vocabulary needed to answer them. It defines the basic electrical quantities — EMF, current, potential difference, power, and energy — introduces the three passive components (resistor, capacitor, inductor), distinguishes DC, AC, and pulsating signals, and develops the ideal and practical source models used in the rest of the book.

1.1 Basic Electrical Quantities

EMF and potential difference

A source such as a cell, battery, or generator supplies energy to electric charges. The energy delivered per unit charge is the **electromotive force** (EMF), measured in volts. EMF is not a mechanical force; the name is historical. If a source supplies energy W joules to a charge Q coulombs,

$$V = \frac{W}{Q} \tag{1.1}$$

One volt equals one joule per coulomb.

The **potential difference** between two points is the difference in electric potential between them. In circuit work, a voltmeter placed across a resistor, lamp, or battery terminal reads the potential difference between those points. EMF refers specifically to the source; potential difference refers to the voltage between any two points in a circuit. The numerical values may coincide in simple problems, but the terms are not interchangeable.

Figure 1.1 contrasts the EMF provided by the source with the potential difference across the lamp and shows the two current conventions.

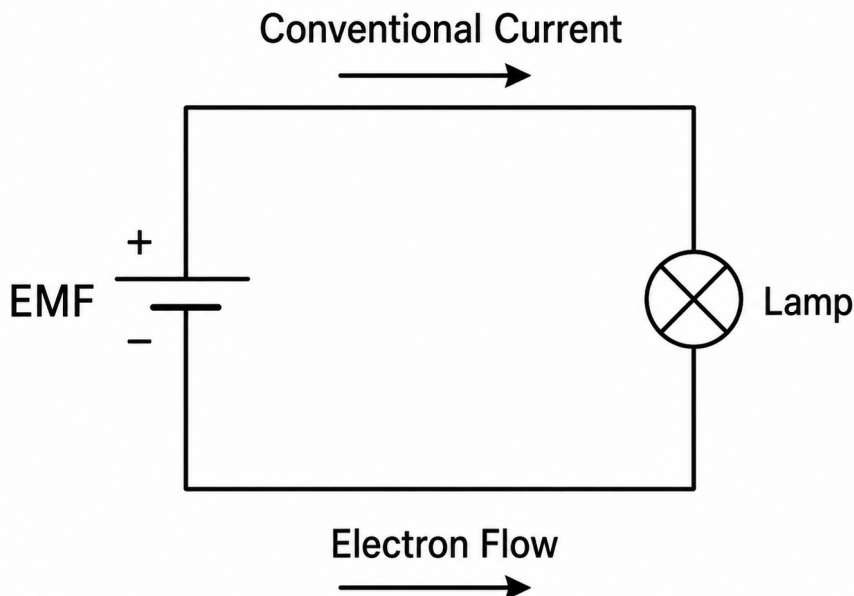


Figure 1: *Figure 1.1: Simple battery-lamp circuit showing source EMF, lamp, conventional current direction, and electron flow.*

Electric current

Current is the flow of electric charge through a conductor. If a charge Q passes a cross-section in time t ,

$$I = \frac{Q}{t} \quad (1.2)$$

One ampere equals one coulomb per second. In metals the carriers are electrons; in electrolytes and semiconductors the carriers differ, but for circuit analysis only the net rate of charge transfer matters.

Power and energy

A lamp glows, a heater warms, and a motor turns because electrical energy is converted to other forms. The rate of this conversion is the **electric power**. For a DC circuit,

$$P = VI \quad (1.3)$$

with P in watts. **Electrical energy** is the time-integral of power. For constant P over time t ,

$$W = Pt = VIt \quad (1.4)$$

In SI, energy is in joules when P is in watts and t in seconds. For longer durations, energy is usually expressed in watt-hour or kilowatt-hour:

$$1 \text{ Wh} = 3600 \text{ J}, \quad 1 \text{ kWh} = 3.6 \times 10^6 \text{ J} \quad (1.5)$$

The kWh on an electricity bill is a unit of energy, not power.

Table 1.1: Basic electrical quantities

Quantity	Symbol	SI unit	Meaning
EMF / voltage	V or $\mathcal{E}$	volt (V)	energy per unit charge
Current	I	ampere (A)	rate of flow of charge
Power	P	watt (W)	rate of energy transfer
Energy	W or E	joule (J)	total electrical work done
Charge	Q	coulomb (C)	quantity of electricity

Worked Example 1.1

A 12 V battery supplies 0.5 A to a lamp for 3 hours. Find the power and the energy consumed.

$$P = VI = 12 \times 0.5 = 6 \text{ W}$$

With $t = 3 \text{ h} = 10,800 \text{ s}$,

$$W = Pt = 6 \times 10,800 = 64,800 \text{ J} = 18 \text{ Wh}$$

Worked Example 1.2

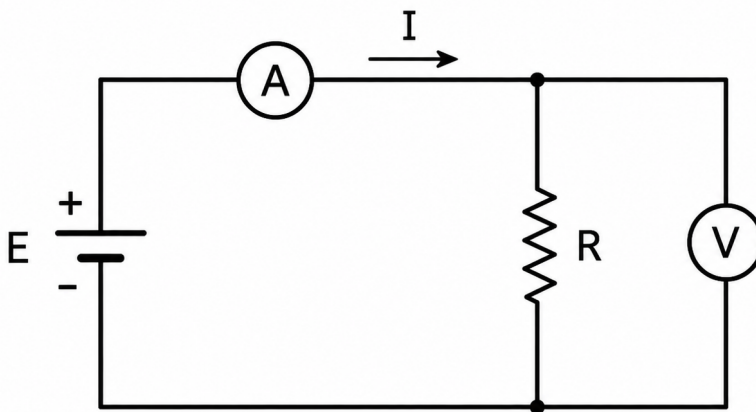
A 230 V table fan draws 0.25 A. Estimate its input power.

$$P = VI = 230 \times 0.25 = 57.5 \text{ W}$$

This is a first estimate. For AC loads the true average power also depends on the power factor, as shown in the AC chapter.

A final caution on measurement: a voltmeter is connected across a component, an ammeter in series with it. Reversing these connections is the most common cause of blown multimeter fuses in the laboratory.

Figure 1.2 shows the correct multimeter connection and the most common wrong arrangement to avoid.



Ammeter in Series
Voltmeter in Parallel

Figure 2: Figure 1.2: Ammeter connected in series and voltmeter connected in parallel across the resistor.

1.2 Passive Components

The resistor, capacitor, and inductor are **passive** components because they do not generate electrical energy. They are the elements used to control current, store energy, and shape signal behaviour in every circuit that follows.

Resistor

A **resistor** opposes current, and its opposition is called **resistance** R , measured in ohms (Ω). For a linear resistor, Ohm's law relates voltage and current:

$$V = IR \quad (1.6)$$

The power dissipated in a resistor can be written in three equivalent forms that follow from Ohm's law together with $P = VI$:

$$P = VI = I^2R = \frac{V^2}{R} \quad (1.7)$$

A resistor converts this power almost entirely into heat. Resistors are used for current limiting, voltage division, transistor biasing, and deliberate heat generation in appliances such as irons and geysers.

Fixed and variable resistors

A **fixed resistor** has a single nominal value set at manufacture. A **variable resistor** can be adjusted during use. The common variable types are the **potentiometer**, a three-terminal device used as an adjustable voltage divider; the **rheostat**, a two-terminal device used for current control; and the **preset** or **trimmer**, a small in-circuit device used for factory or field calibration. Fixed resistors dominate in practical circuits; variable resistors are reserved for functions

that genuinely need manual adjustment, such as volume, tuning, sensitivity, or calibration.

Ratings and tolerance

A resistor is selected for three parameters: resistance value, **power rating**, and **tolerance**. Common small through-hole power ratings are 0.125 W, 0.25 W, 0.5 W, and 1 W; a comfortable margin between calculated dissipation and rated dissipation extends service life. Tolerance specifies the maximum allowed deviation from the nominal value: a 1 k Ω resistor at $\pm 5\%$ may measure between 950 Ω and 1050 Ω , while a 10 k Ω resistor at $\pm 1\%$ lies between 9.9 k Ω and 10.1 k Ω . Standard colour-band coding identifies nominal value and tolerance at a glance; a worked reading of the code is given at the end of this chapter.

Temperature effect

The resistance of a real resistor is not strictly constant with temperature. In most metallic resistors, resistance rises slightly as temperature rises, because increased lattice vibration scatters the conduction electrons more strongly. This behaviour is summarised by the temperature coefficient of resistance. It is negligible in many low-power calculations but matters for precision instrumentation, high-current paths where self-heating is significant, and circuits required to be stable over a wide ambient range.

Voltage divider

Two resistors in series across a source act as a **voltage divider**. With R_1 and R_2 in series across a source V_s , the same current

$$I = \frac{V_s}{R_1 + R_2}$$

flows through both, so the voltage across R_2 is

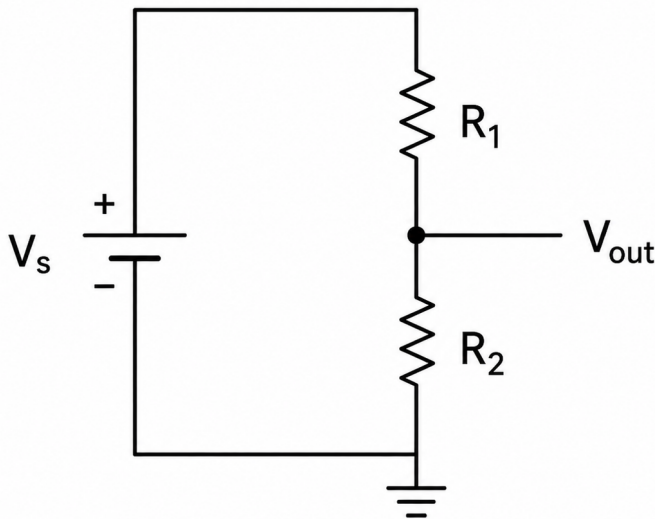
$$V_{out} = \frac{R_2}{R_1 + R_2} V_s \quad (1.8)$$

For example, with $V_s = 12 \text{ V}$, $R_1 = 2 \text{ k}\Omega$, and $R_2 = 1 \text{ k}\Omega$,

$$V_{out} = \frac{1}{2 + 1} \times 12 = 4 \text{ V}$$

The divider relation is one of the most frequently used small formulas in electronics, appearing in transistor biasing, sensor reference networks, and signal conditioning.

Figure 1.3 shows the divider tap and the reference to ground explicitly.



Output taken with respect to ground

Figure 3: Figure 1.3: Voltage divider with output V_{out} taken from the junction of R_1 and R_2 with respect to ground.

Worked Example 1.3

A 10 V source is connected across a 2 kΩ resistor. Find the current and the power dissipated.

$$I = \frac{V}{R} = \frac{10}{2000} = 5 \text{ mA}$$

$$P = VI = 10 \times 0.005 = 0.05 \text{ W}$$

A 0.25 W resistor is a safe practical choice for this dissipation.

Capacitor

A **capacitor** stores energy in the electric field between two conducting surfaces separated by a dielectric. The fundamental relation is

$$C = \frac{Q}{V} \quad (1.9)$$

where C is in farads, Q in coulombs, and V is the voltage across the capacitor. Capacitance measures the charge stored per volt. The energy stored is

$$W_C = \frac{1}{2}CV^2 \quad (1.10)$$

Figure 1.4 links the field-based picture of capacitance with the polarity marking that matters for electrolytic capacitors.

When a capacitor is connected to a DC source through a resistor, the charging current is maximum at the first instant and falls as the capacitor voltage rises. The process is governed by the **time constant**

$$\tau = RC \quad (1.11)$$

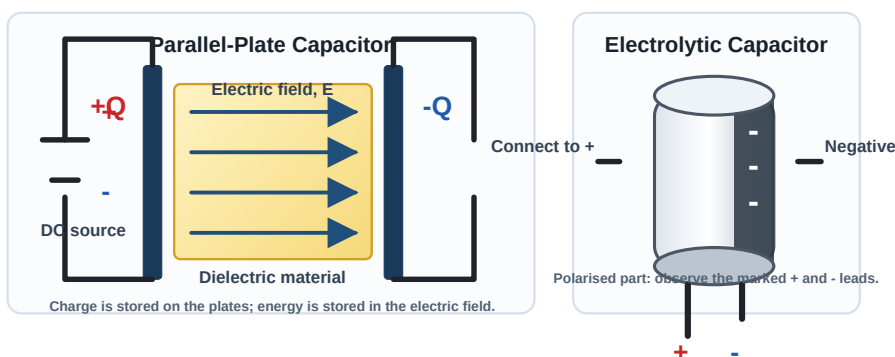


Figure 4: Figure 1.4: Parallel-plate capacitor showing dielectric and electric field lines beside an electrolytic capacitor polarity sketch.

with R in ohms, C in farads, and τ in seconds. Larger R or larger C gives slower charging and discharging; after roughly five time constants the transient is practically complete. A fully charged capacitor in an ideal DC circuit passes no further current and behaves as an open circuit. In AC circuits the voltage changes continuously, so a capacitor repeatedly charges and discharges, and current flows in the branch at all times.

Figure 1.5 shows the charging and discharging transients, with τ and 5τ marked on the time axis.

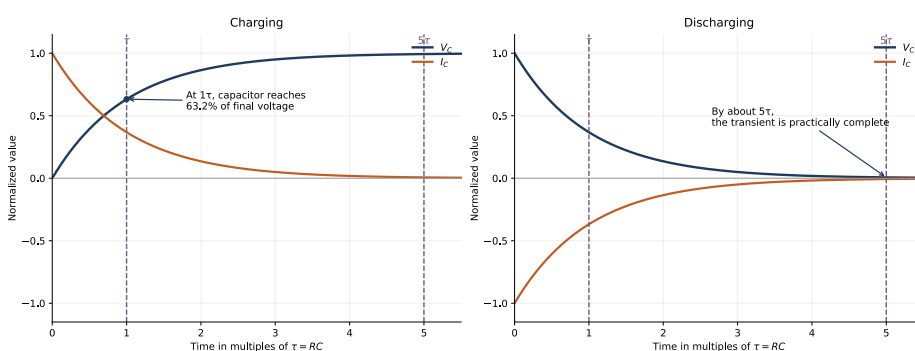


Figure 5: Figure 1.5: RC charging and discharging curves for capacitor voltage and current, with tau and five tau indicated.

Typical applications include smoothing of rectifier outputs, AC

coupling between stages, bypassing of supply noise, timing networks, and power-factor correction.

Capacitor types and polarity

Different dielectrics suit different duties. **Ceramic** capacitors are small, non-polarised, and dominate in decoupling and high-frequency bypassing. **Electrolytic** capacitors offer high capacitance in a small volume but are polarised and must be connected with the correct polarity; reverse connection beyond the rated reverse voltage can cause heating, leakage, or violent failure. **Film** capacitors are stable and reliable, favoured in AC, audio, timing, and motor-run applications. **Mica** and other specialist capacitors are used where stability or high-frequency performance is critical. Selection therefore depends on voltage rating, polarity, tolerance, intended AC or DC duty, and the physical and thermal environment — not on capacitance value alone.

Worked Example 1.4

A 1000 μF capacitor is charged to 12 V. Find the energy stored.

$$C = 1000 \times 10^{-6} \text{ F} = 10^{-3} \text{ F}$$

$$W_C = \frac{1}{2}CV^2 = \frac{1}{2} \times 10^{-3} \times 144 = 0.072 \text{ J}$$

The value is small compared with battery energy but is routinely exploited in smoothing and short-term hold-up applications.

Inductor

An **inductor** is a coil that stores energy in the magnetic field set up by its current. The voltage across an ideal inductor is proportional to the rate of change of current:

$$v = L \frac{di}{dt} \quad (1.12)$$

with L in henrys. The inductor therefore opposes changes in current: a rapid attempt to change i produces a large opposing v . The energy stored is

$$W_L = \frac{1}{2}LI^2 \quad (1.13)$$

Inductance depends on the number of turns, coil geometry, magnetic path length, and core material. **Air-core** inductors avoid core losses and suit higher-frequency use. **Iron-core** inductors give high inductance per turn at power frequencies. **Ferrite-core** inductors dominate in modern electronic equipment because ferrites combine useful permeability with low high-frequency loss.

A practical coil is made of wire with finite resistance R_w , and so dissipates $P_{cu} = I^2 R_w$ in copper loss. This is why an apparently inductive coil still warms in service. In a magnetic-core inductor, increasing current eventually drives the core into **saturation**, where flux no longer rises in proportion to current. Inductance falls, current may rise more steeply than the design predicts, and additional heating results. Saturation is a central concern for power inductors, transformers, relays, and magnetic actuators; Unit 2 examines its magnetic origin.

Figure 1.6 combines the magnetic-field picture of a coil with the

idea of back EMF during current change.

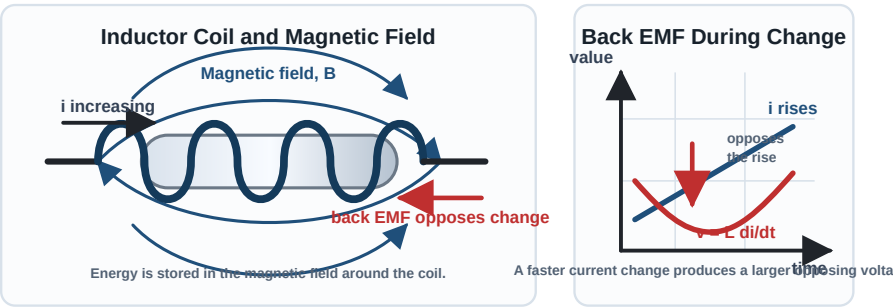


Figure 6: Figure 1.6: Inductor coil with magnetic field lines and a back-EMF sketch showing opposition to rising current.

Worked Example 1.5

An inductor of 20 mH carries a current of 2 A. Find the stored energy.

$$W_L = \frac{1}{2}LI^2 = \frac{1}{2} \times 0.02 \times 4 = 0.04 \text{ J}$$

Table 1.2: Comparison of the three passive components

Component	Main effect	Unit	Stores energy in	Steady-DC	Common use
				behaviour	
Resistor	opposes current	Ω	none (ideally)	$V = IR$	current limiting, voltage division

Component	Main effect	Unit	Stores energy in	Steady-DC behaviour		Common use
Capacitor	opposes change in voltage	F	electric field	open circuit once charged		filtering, coupling, timing
Inductor	opposes change in current	H	magnetic field	short circuit once settled		chokes, coils, filters

Resistor values can be read from colour bands and verified with a multimeter; capacitance and inductance are measured with an LCR meter, which is why the LCR meter is a standard instrument in the FEEE laboratory.

1.3 Signal Waveforms

A **waveform** is the pattern by which a voltage or current varies with time. A battery produces nearly steady DC; a rectifier output before filtering produces pulsating DC; the mains supply produces AC; a switching transient produces a single non-repeating event.

DC, pulsating DC, and AC

A **DC signal** has a single polarity. Steady DC has constant magnitude as well; pulsating DC varies in magnitude but never reverses

polarity. An **AC signal** changes both in magnitude and polarity, reversing direction periodically. The Indian single-phase domestic supply is 230 V at 50 Hz. The distinction matters because a rectifier output before the smoothing capacitor is neither pure AC nor steady DC — it is pulsating DC.

Periodic and non-periodic waveforms

A waveform is **periodic** if it repeats after a fixed time interval T , called the **time period**. Its **frequency** is

$$f = \frac{1}{T} \quad (1.14)$$

measured in hertz. At 50 Hz the period is $T = 1/50 = 20$ ms, so one mains cycle occupies 20 ms. A **non-periodic** waveform has no regular repetition — a one-shot discharge, a switching transient, or a random sensor event are typical examples.

Amplitude, peak-to-peak value, and DC offset

The **amplitude** of a waveform is its maximum departure from the reference level. A sine wave swinging between +10 V and −10 V has an amplitude of 10 V and a peak-to-peak value of 20 V. The **peak-to-peak value** is always the total swing between the extreme values, regardless of whether the waveform crosses zero: a signal that varies between 2 V and 8 V has $V_{pp} = 6$ V. The **DC offset** is the displacement of the waveform's average value from zero; the 2–8 V signal above has an offset of 5 V.

An oscilloscope displays voltage against time, with the vertical scale set in volts per division and the horizontal scale in seconds per division. Both scale settings must be read before interpreting any trace, as otherwise a correct waveform can appear misleading.

Waveform families

A **sine wave** varies smoothly and periodically. It is central to power engineering because rotating alternators produce it naturally and because AC circuit analysis takes its simplest form for sinusoids. A **square wave** alternates sharply between two levels and is the workhorse of digital circuits and clock signals. A **triangular wave** rises and falls linearly and is used in timing and waveform generation. A **sawtooth wave** rises linearly and then resets rapidly, appearing in scan and sweep circuits. A **pulse** sits at one level for a defined time and then briefly switches to another; repeated pulses drive digital control, triggering, communication, and switched power conversion.

Figure 1.7 compares common waveform families and marks amplitude, peak-to-peak value, period, frequency, DC offset, and duty cycle on representative traces.

Duty cycle

For pulse and square waveforms, the **duty cycle** is the fraction of one period spent in the high state,

$$D = \frac{t_{on}}{T} \times 100 \% \quad (1.15)$$

A pulse high for 2 ms in every 10 ms period has a duty cycle of 20 %. Duty cycle is central to digital signalling, motor speed control, DC–DC conversion, and pulse-width-modulated power delivery.

Worked Example 1.6

A waveform repeats every 5 ms. Find its frequency.

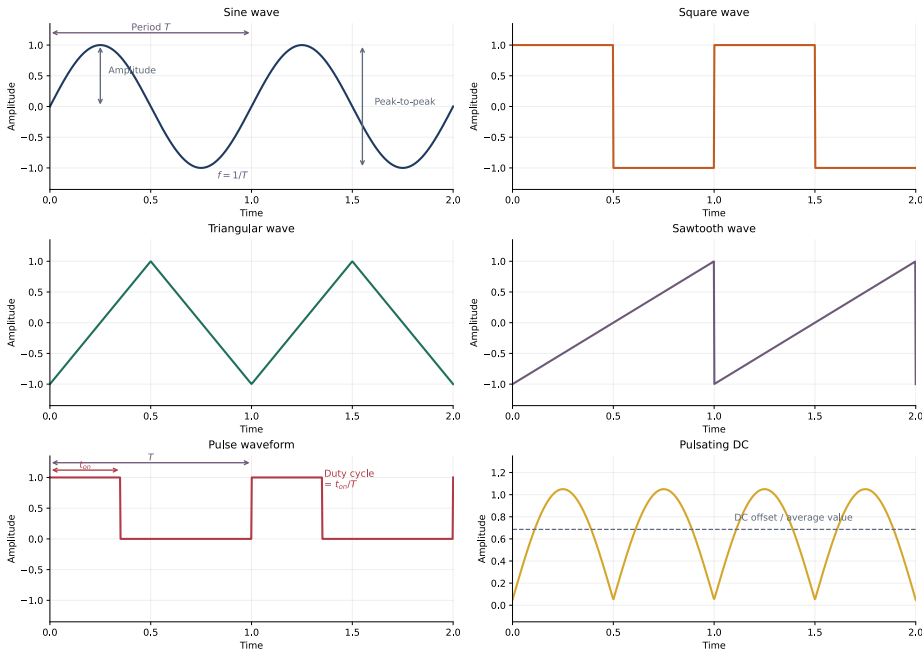


Figure 7: Figure 1.7: Common waveform families including sine, square, triangular, sawtooth, pulse, and pulsating DC with amplitude, peak-to-peak value, period, frequency, DC offset, and duty-cycle annotations.

$$f = \frac{1}{T} = \frac{1}{5 \times 10^{-3}} = 200 \text{ Hz}$$

Worked Example 1.7

A pulse waveform has a period of 8 ms and a high-state time of 3 ms. Find the duty cycle.

$$D = \frac{3}{8} \times 100 \% = 37.5 \%$$

Worked Example 1.8

A DSO is set to 5 ms/div and one complete cycle occupies four horizontal divisions. Find the frequency.

$$T = 4 \times 5 \text{ ms} = 20 \text{ ms}, \quad f = \frac{1}{0.02} = 50 \text{ Hz}$$

Table 1.3: Simple waveform classification

Signal type	Repeats regularly?	Polarity	Example
Steady DC	no time variation	one polarity	dry cell, regulated 5 V supply
Pulsating DC	often periodic	one polarity	rectifier output before smoothing
AC periodic	yes	changes polarity	230 V, 50 Hz mains
Non-periodic	no	may vary	transient pulse, switching event

Not every AC waveform is sinusoidal — a square wave that alternates in polarity is equally AC — and not every DC signal is flat, as pulsating DC demonstrates. Frequency and amplitude describe independent properties: frequency is how often the pattern repeats, amplitude is how large each cycle is.

1.4 Sources

Every practical circuit needs an energy source — a battery, bench supply, rectifier, generator, solar module, or signal generator. For analysis these are replaced by idealised source models, because no real source is perfect and explicit modelling of the imperfection is essential when the imperfection matters.

Figure 1.8 summarises the ideal and practical voltage-source and current-source models used throughout this section.

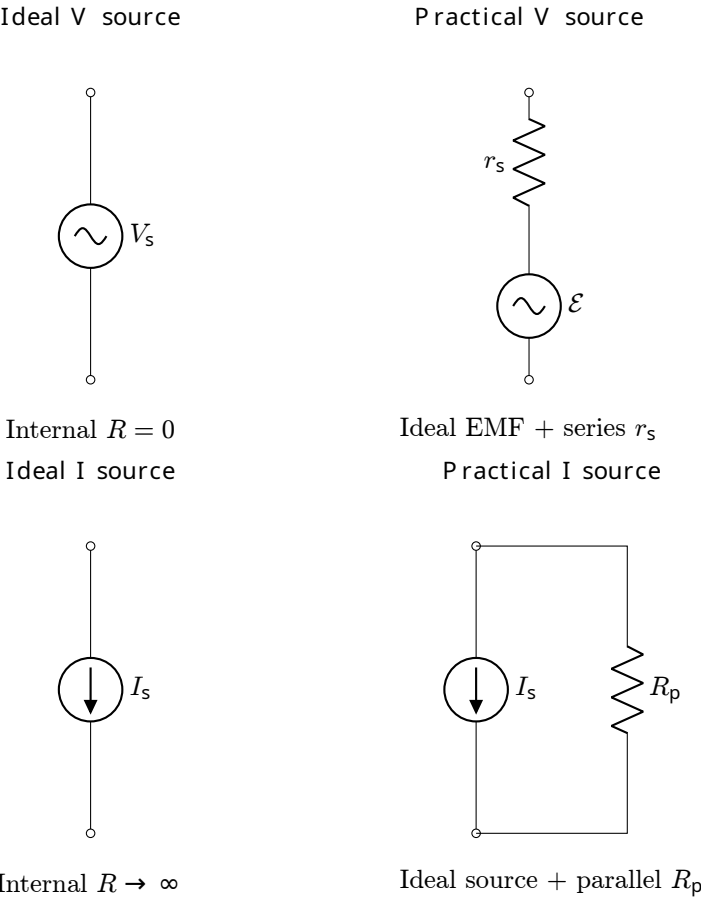


Figure 8: Figure 1.8: Side-by-side ideal and practical voltage-source and current-source equivalent circuits.

An **ideal voltage source** holds its terminal voltage constant for any load current; its internal resistance is zero. A real source always has a finite internal resistance r_s , and is modelled as an ideal EMF $\mathcal{E}$ in series with r_s . For a load current I ,

$$V = \mathcal{E} - I r_s \quad (1.16)$$

The drop $I r_s$ inside the source is why a battery terminal voltage sags when the load is heavy.

Terminal characteristic

The **open-circuit voltage** is the terminal voltage when no load current is drawn; for the practical voltage source this equals $\mathcal{E}$. The **short-circuit current** is the current that would flow if the terminals were bridged by an ideal short:

$$I_{sc} = \frac{\mathcal{E}}{r_s} \quad (1.17)$$

This is a fault condition, not a design point; in real sources it is limited by internal construction, wiring resistance, protection devices, and thermal response. Equation (1.16) plots as a straight line on axes of V against I : terminal voltage $\mathcal{E}$ at $I = 0$, falling with slope $-r_s$ and reaching zero at $I = I_{sc}$. This **terminal characteristic** captures the no-load, loaded, and fault behaviour in a single diagram.

Figure 1.9 plots the terminal characteristic of a practical voltage source.

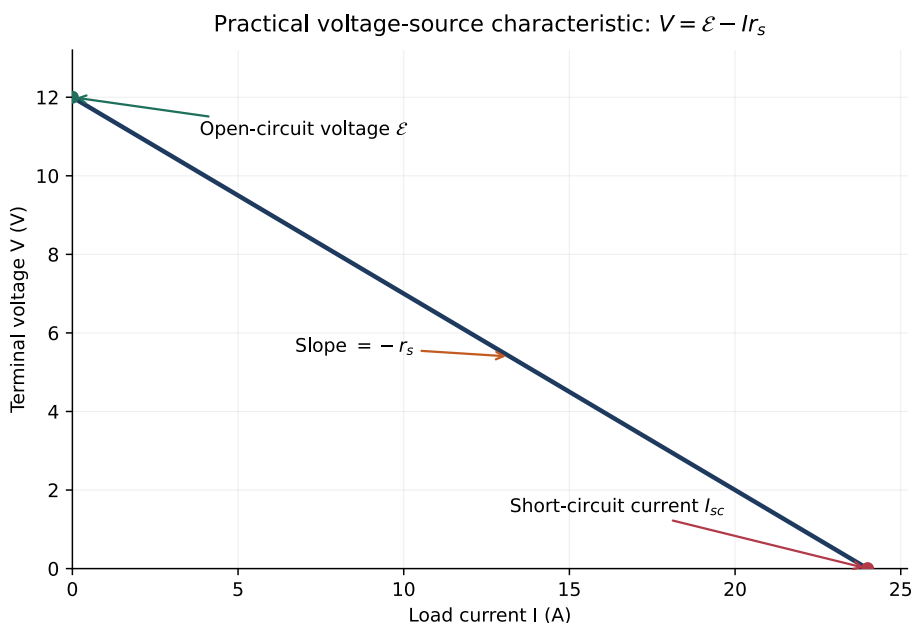


Figure 9: Figure 1.9: Terminal characteristic $V = \mathcal{E} - I r_s$ of a practical voltage source, showing open-circuit voltage, short-circuit current, and the slope $-r_s$.

Load regulation

Load regulation describes how much the terminal voltage changes between no-load and full-load. A well-regulated source holds its voltage nearly constant as the load current varies; a poorly regulated source shows a pronounced droop. The effect is important in batteries under heavy discharge, bench DC supplies, adapters, chargers, and any regulated electronic power supply. A battery that reads 12 V on open circuit may deliver noticeably less when supplying a motor — that is load regulation in action.

Figure 1.10 visualises the voltage droop from no-load to full-load.

Worked Example 1.9

A battery of EMF 12 V and internal resistance $0.5 \, \Omega$ supplies 2 A to a load. Find the terminal voltage.

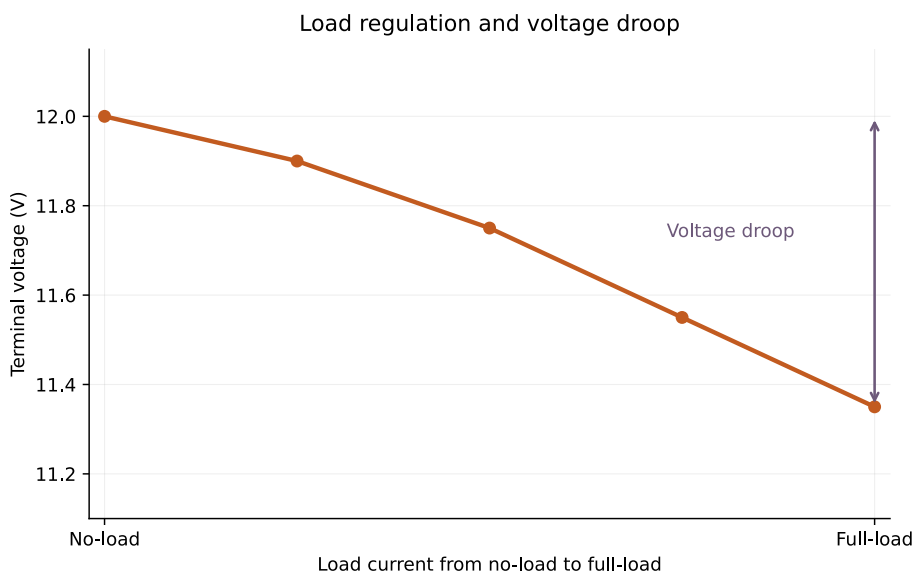


Figure 10: Figure 1.10: Small voltage-droop graph showing terminal voltage falling from no-load to full-load.

$$V = 12 - (2 \times 0.5) = 11 \text{ V}$$

Ideal and practical current sources

An **ideal current source** holds its output current constant for any terminal voltage; its internal resistance is infinite. A practical current source is modelled as an ideal current source in parallel with a large internal resistance R_p . The parallel placement is the correct choice because a large parallel resistance diverts very little of the source current away from the load. Current sources appear naturally in transistor bias networks, current mirrors, and sensor interfaces, even when no explicit current-source symbol appears in the final schematic.

Worked Example 1.10

An ideal 10 mA current source drives a 500 Ω load. Find the load voltage.

$$V = IR = 0.01 \times 500 = 5 \text{ V}$$

The current is fixed by the source; the voltage adjusts to whatever the load requires.

Source transformation

A practical voltage source can be replaced by an equivalent practical current source and vice versa. If a voltage source $\mathcal{E}$ is in series with resistance R , the equivalent current-source form is

$$I_s = \frac{\mathcal{E}}{R} \quad (1.18)$$

in parallel with the same resistance R . Conversely,

$$\mathcal{E} = I_s R \quad (1.19)$$

The series resistance in one form becomes the parallel resistance in the other; its value is unchanged. **Source transformation** is a mathematical manoeuvre on the model, not a physical change to the source: the external behaviour at the terminals is identical in both forms, and one form is often easier to analyse than the other. Figure 1.11 shows the voltage-source and current-source forms feeding the same load.

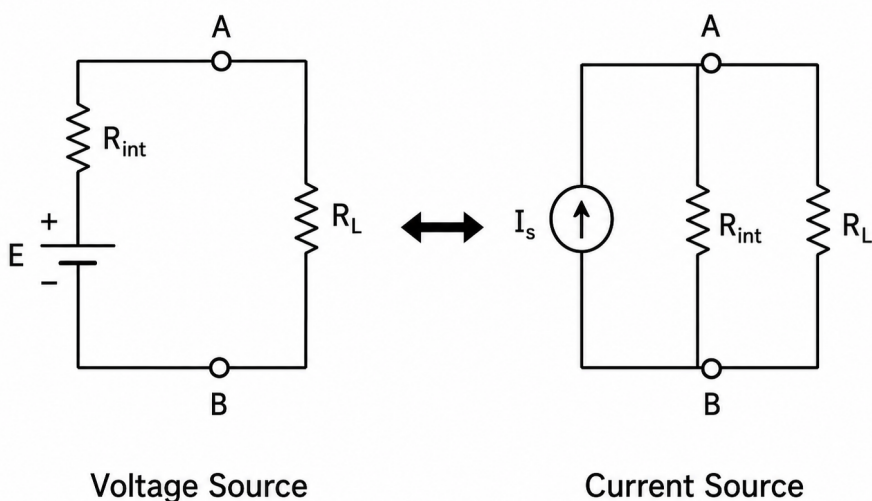


Figure 11: Figure 1.11: Equivalent voltage-source and current-source forms connected to the same load.

Worked Example 1.11

Transform a 12 V source in series with 6 Ω into its equivalent current-source form.

$$I_s = \frac{12}{6} = 2 \text{ A}$$

The equivalent is a 2 A current source in parallel with 6 Ω .

Worked Example 1.12

Connect the source of Example 1.11 to a 3 Ω load and find the load current by both forms.

Voltage-source form. Total loop resistance is $6 + 3 = 9 \text{ } \Omega$, so

$$I = \frac{12}{9} = 1.333 \text{ A}$$

Current-source form. The $6\ \Omega$ internal and $3\ \Omega$ load appear in parallel:

$$R_{eq} = \frac{6 \times 3}{6 + 3} = 2\ \Omega, \quad V = I_s R_{eq} = 2 \times 2 = 4\ \text{V}$$

$$I_L = \frac{V}{R_L} = \frac{4}{3} = 1.333\ \text{A}$$

Both forms give the same load current, confirming the equivalence.

Table 1.4: Ideal and practical source models

Source type	Ideal internal resistance	Practical form	Intended behaviour
Voltage source	$0\ \Omega$	ideal EMF in series with small r_s	constant terminal voltage
Current source	infinite	ideal current source in parallel with large R_p	constant output current

An ideal voltage source is defined by constancy of voltage, not by the voltage being large; likewise an ideal current source is defined by constancy of current. No real source maintains its rated value under every condition, which is the practical reason internal resistance appears in the model at all.

Worked Interpretation Exercise

Reading the Resistor Colour Code

The standard four-band resistor colour code encodes two significant digits, a decimal multiplier, and a tolerance. Consider a resistor with bands **Brown – Black – Red – Gold**.

Figure 1.12 summarises the four-band resistor colour code and shows a worked colour-band example.

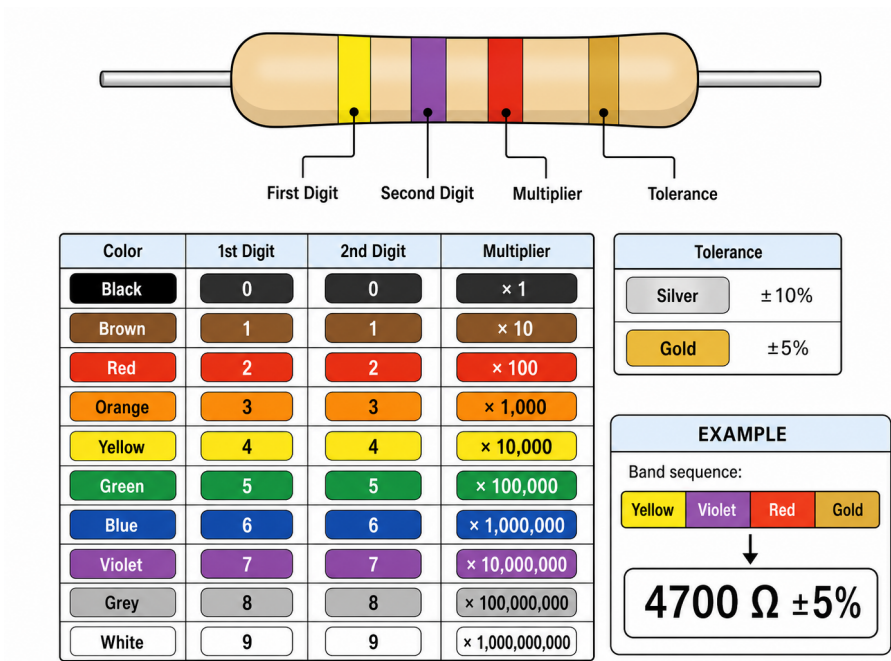


Figure 12: Figure 1.12: Four-band resistor colour-code chart with digit, multiplier, tolerance, and worked example.

- Brown gives the first digit, 1.
- Black gives the second digit, 0.
- Red is the multiplier, 10^2 .
- Gold is the tolerance, $\pm 5\%$.

The nominal value is therefore

$$R = 10 \times 10^2 = 1000 \, \Omega = 1 \, \text{k}\Omega$$

A tolerance of $\pm 5 \%$ allows an actual value in the range

$$R_{\min} = 1000 - 50 = 950 \, \Omega, \quad R_{\max} = 1000 + 50 = 1050 \, \Omega$$

A multimeter reading of $0.99 \, \text{k}\Omega$ on the resistor, measured out of circuit, is consistent with the code. This exercise links component marking, nominal value, manufacturing tolerance, and measured value in a single practical task, and is a laboratory skill worth developing early.

Further Reading

1. Charles K. Alexander and Matthew N. O. Sadiku, *Fundamentals of Electric Circuits* — widely used circuit-analysis text for sources, passive elements, and basic circuit laws.
2. James W. Nilsson and Susan A. Riedel, *Electric Circuits* — standard undergraduate text with strong treatment of voltage, current, power, source models, and first-order circuits.
3. Edward Hughes, John Hiley, Keith Brown, and Ian McKenzie Smith, *Electrical and Electronic Technology* — popular introductory text for electrical quantities, components, measurements, and practical technology.
4. V. K. Mehta and Rohit Mehta, *Principles of Electrical Engineering and Electronics* — widely used introductory text for circuit quantities, passive elements, and practical electrical concepts.

Magnetic Circuits and Electromagnetic Induction

{Overview

} This unit develops magnetic-circuit ideas and electromagnetic induction, connecting field concepts to induced EMF, energy transfer, and the operating principles behind transformers and machines.

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}

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Magnetic Circuit Fundamentals 2.2 Electromagnetic Induction
2.3 Direction Rules

Unit 2: Magnetic Circuits and Electromagnetic Induction

A current flowing through a coil produces a magnetic effect, and that effect is what makes relays pull, transformers transfer energy, motors rotate, and generators deliver voltage. This chapter develops the two ideas behind those devices: the magnetic circuit, with its quantities of magnetomotive force, flux, permeability, reluctance, and core loss; and electromagnetic induction, which links a changing flux to an induced emf and leads directly to self-inductance, mutual inductance, and Fleming's rules.

2.1 Magnetic Circuit Fundamentals

A current in a straight wire sets up a magnetic field around it. The same current in a coil wound on an iron core produces a much stronger field, because iron and steel carry magnetic flux far more easily than air. Engineers describe the preferred path of that flux through iron as a **magnetic circuit**. The name is an analogy for calculation, not a claim that flux is a flowing substance in the sense that current is a flow of charge.

Magnetomotive force

The cause of flux in a magnetic circuit is the **magnetomotive force** (mmf). It is not a mechanical force but the magnetic driving effect

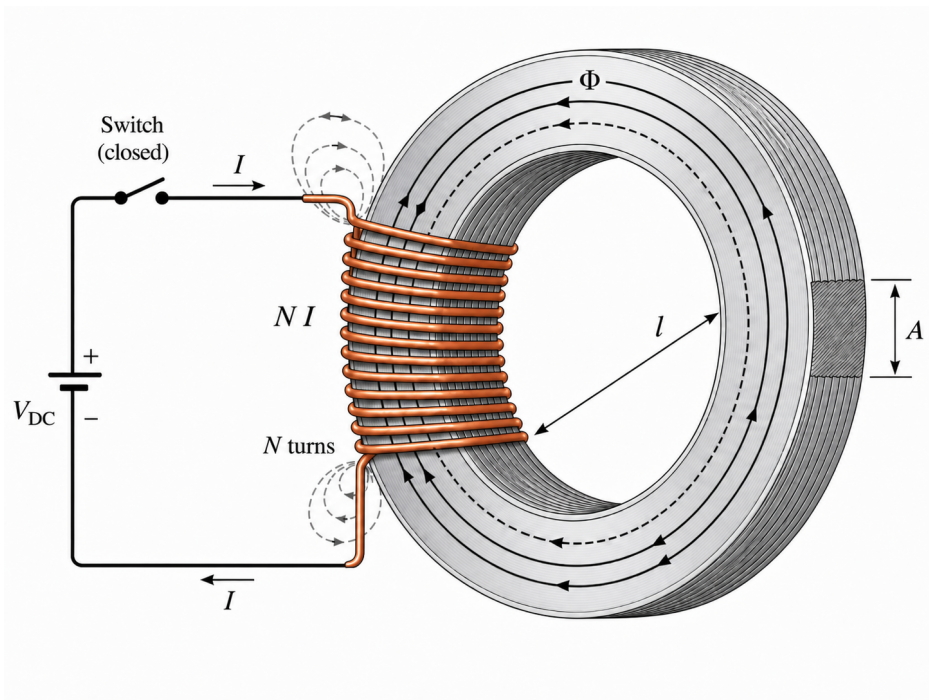


Figure 13: Figure 2.1: Toroidal magnetic circuit showing coil current, magnetomotive force, core flux path, and leakage flux.

produced by a current-carrying coil. For a coil of N turns carrying current I ,

$$\mathcal{F} = NI \quad (2.1)$$

where $\mathcal{F}$ is in ampere-turns. The mmf scales linearly with both N and I .

Magnetic flux and flux density

The magnetic effect set up in the core is described by the **magnetic flux** Φ , measured in webers (Wb). Flux tells how much magnetic field passes through a given cross-section. If the flux is spread over a cross-sectional area A , the **magnetic flux density** is

$$B = \frac{\Phi}{A} \quad (2.2)$$

with B in tesla (T). In machine and transformer cores, flux density is the quantity to watch, because high B drives the core toward saturation.

Magnetic field intensity and permeability

The magnetizing effort per unit length of the magnetic path is the **magnetic field intensity**,

$$H = \frac{NI}{l} \quad (2.3)$$

in ampere-turns per metre, where l is the mean length of the magnetic path. Flux density and field intensity are related by

$$B = \mu H \quad (2.4)$$

where μ is the **permeability** of the material, a measure of how easily the material supports flux. It is conventionally written as

$$\mu = \mu_0 \mu_r \quad (2.5)$$

in which μ_0 is the permeability of free space and μ_r is the **relative permeability** of the material. Air has $\mu_r \approx 1$; iron and silicon steel have values orders of magnitude larger, though μ_r itself varies with the level of magnetization. This is why an iron-cored coil produces so much more flux than the same coil in air.

Reluctance

Just as resistance opposes current in an electric circuit, **reluctance** $\mathcal{R}$ opposes flux in a magnetic circuit:

$$\mathcal{R} = \frac{l}{\mu A} \quad (2.6)$$

Once reluctance is known, the flux follows a circuit-like relation,

$$\Phi = \frac{\mathcal{F}}{\mathcal{R}} \quad (2.7)$$

often compared to Ohm's law. The analogy is useful for first calculations but must be used with care: resistance is approximately constant, whereas reluctance varies because permeability itself varies with magnetization.

Table 2.1: Electric and magnetic circuit analogy

Electric circuit	Symbol	Magnetic circuit	Symbol
Electromotive force	V	Magnetomotive force	$\mathcal{F}$
Current	I	Flux	Φ
Resistance	R	Reluctance	$\mathcal{R}$
Conductance	$1/R$	Permeance	$1/\mathcal{R}$
Current density	J	Flux density	B

The analogy is not exact. Electric current is a real flow of charge; magnetic flux is not. A resistor dissipates power whenever current flows; a magnetic core dissipates energy mainly when flux changes, through hysteresis and eddy-current losses. And because permeability is not constant, magnetic circuits are generally nonlinear.

Series magnetic paths and air gaps

Practical magnetic circuits are rarely a single uniform path. A transformer core has corners and limbs, an electromagnet has an

iron path broken by a working-face air gap, and a relay has a path through core, armature, and small gaps. For sections in series the reluctances add,

$$\mathcal{R}_{total} = \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3 + \dots$$

with each section found from $\mathcal{R} = l/(\mu A)$.

An **air gap** often dominates the total reluctance even when it is physically short, because $\mu_r \approx 1$ in air against μ_r of many hundreds or thousands in iron. For a given flux density B , a small value of H is enough inside the core but a much larger H is required across the gap. The current the coil must draw to establish a required flux is therefore set largely by the gap, which is why relay, electromagnet, and inductor designers treat gap length as a primary design variable: small mechanical changes in the gap produce large changes in flux and current.

Leakage flux and leakage factor

In an ideal core all flux would stay in the intended path. Real cores allow some flux to spread into the surrounding air; this part is called **leakage flux**. If Φ_u is the useful flux and Φ_t the total flux produced, the **leakage factor** is

$$\lambda = \frac{\Phi_t}{\Phi_u} \quad (2.8)$$

and is always greater than 1. Leakage flux influences coupling, voltage regulation, stray fields, and unwanted heating of nearby metal parts.

Worked Example 2.1

A toroidal iron core has mean magnetic path length $l = 0.4$ m and cross-sectional area $A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$. A coil of 500 turns carries a current of 0.4 A. At this operating point the core has relative permeability 800. Find the mmf, reluctance, flux, and flux density.

$$\mathcal{F} = NI = 500 \times 0.4 = 200 \text{ A-turns}$$

$$\mu = \mu_0 \mu_r = 4\pi \times 10^{-7} \times 800 \approx 1.005 \times 10^{-3} \text{ H/m}$$

$$\mathcal{R} = \frac{l}{\mu A} = \frac{0.4}{(1.005 \times 10^{-3})(4 \times 10^{-4})} \approx 9.95 \times 10^5 \text{ A-turns/Wb}$$

$$\Phi = \frac{\mathcal{F}}{\mathcal{R}} = \frac{200}{9.95 \times 10^5} \approx 2.01 \times 10^{-4} \text{ Wb}$$

$$B = \frac{\Phi}{A} = \frac{2.01 \times 10^{-4}}{4 \times 10^{-4}} \approx 0.503 \text{ T}$$

Worked Example 2.2

The total flux produced in a magnetic system is 20 mWb; the useful flux linking the required part of the system is 18 mWb. Find the leakage factor.

$$\lambda = \frac{\Phi_t}{\Phi_u} = \frac{20}{18} = 1.11$$

The total flux exceeds the useful flux by about 11%.

Worked Example 2.3

A magnetic circuit consists of an iron path and a single air gap in series. The iron path has length 0.30 m, cross-sectional area $5 \times 10^{-4} \text{ m}^2$, and relative permeability 1000 at the operating point. The air gap is 1 mm long with the same area. A 400-turn coil carries 0.5 A. Find the iron-path reluctance, the gap reluctance, the total reluctance, and the flux.

$$\mathcal{F} = NI = 400 \times 0.5 = 200 \text{ A-turns}$$

$$\mu_{\text{iron}} = 4\pi \times 10^{-7} \times 1000 \approx 1.257 \times 10^{-3} \text{ H/m}$$

$$\mathcal{R}_{\text{iron}} = \frac{0.30}{(1.257 \times 10^{-3})(5 \times 10^{-4})} \approx 4.77 \times 10^5 \text{ A-turns/Wb}$$

$$\mathcal{R}_{\text{gap}} = \frac{0.001}{(4\pi \times 10^{-7})(5 \times 10^{-4})} \approx 1.59 \times 10^6 \text{ A-turns/Wb}$$

$$\mathcal{R}_{\text{total}} = 4.77 \times 10^5 + 1.59 \times 10^6 \approx 2.07 \times 10^6 \text{ A-turns/Wb}$$

$$\Phi = \frac{\mathcal{F}}{\mathcal{R}_{\text{total}}} = \frac{200}{2.07 \times 10^6} \approx 9.66 \times 10^{-5} \text{ Wb}$$

The 1 mm gap contributes more than three times the reluctance of the 0.30 m iron path.

The B-H curve and saturation

If the current in a magnetizing coil is increased gradually, H rises linearly with current but B does not rise in fixed proportion. A plot of B against H — the **B-H curve** — rises steeply at first, then bends, and finally flattens as the material approaches **saturation**. In saturation, most magnetic domains are already aligned with the applied field, so further increases in H produce only small increases in B . A transformer or inductor core driven far into saturation draws sharply rising current and heats rapidly.

The first rise of B from an unmagnetized state is the **initial magnetization curve**. Along it, domains progressively align with the applied field, which is why a ferromagnetic core delivers much higher flux density than air for the same mmf. Materials differ sharply in how they follow this curve:

- **Soft magnetic materials** (silicon steel, soft iron) magnetize easily and have narrow hysteresis loops, suiting them to transformers, inductors, relays, and machines.
- **Hard magnetic materials** retain magnetism strongly and have wide loops, suiting them to permanent magnets.
- **Ferrites** combine magnetic behaviour with high electrical resistivity, reducing eddy-current loss at higher frequencies.

Hysteresis loop

When a ferromagnetic sample is magnetized first in one direction and then reversed, B does not retrace the initial curve. Instead the (B, H) trajectory forms a closed **hysteresis loop**; the term means that B lags H . Two quantities are read directly from the loop:

- **Retentivity** (residual flux density): the value of B remaining

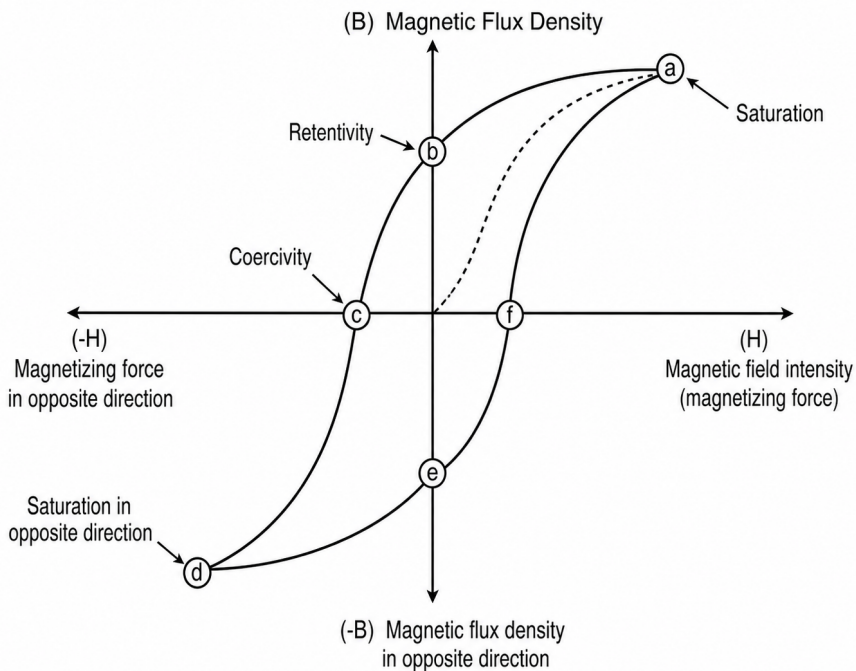


Figure 14: Figure 2.2: B-H curve and hysteresis loop showing saturation, retentivity, and coercivity.

when H is reduced to zero.

- **Coercivity** (coercive force): the reverse value of H required to drive B back to zero.

Soft materials used in transformer cores show narrow loops; hard materials used in permanent magnets show wide ones.

Core losses

The area enclosed by the hysteresis loop equals the energy dissipated per cycle per unit volume of core material. This energy appears as heat, and because the loss is per cycle, the **hysteresis loss** rises with frequency. Choice of a soft magnetic material such as silicon steel keeps the loop narrow and the loss small at power frequencies.

A changing flux also induces circulating currents inside the con-

ducting core itself. These **eddy currents** dissipate I^2R heat in the core. Eddy-current loss is controlled by:

- building the core from thin insulated laminations rather than one solid block, which interrupts the eddy-current paths;
- using high-resistivity magnetic materials; and
- using ferrites at higher frequencies, where laminated steel becomes ineffective.

Hysteresis loss depends on repeated magnetization of the material; eddy-current loss depends on induced currents in the conducting core. Both appear together in AC magnetic equipment and both grow with frequency, which is why transformer cores, motor stators, and machine parts are laminated rather than solid.

2.2 Electromagnetic Induction

Place a coil near a stationary magnet and nothing appears at the terminals. Move the magnet toward the coil and a voltage appears; pull it away and the voltage reverses; hold the magnet still again and the voltage vanishes. This is the central fact of **electromagnetic induction**: an emf is induced in a conductor or coil whenever the magnetic flux linking it changes. Generators, transformers, ignition coils, inductive sensors, chokes, and most switched-mode power converters depend on it.

Faraday's laws

Faraday's experimental results are summarized in two statements:

1. Whenever the magnetic flux linking a circuit changes, an emf is induced in the circuit.
2. The magnitude of the induced emf is proportional to the rate

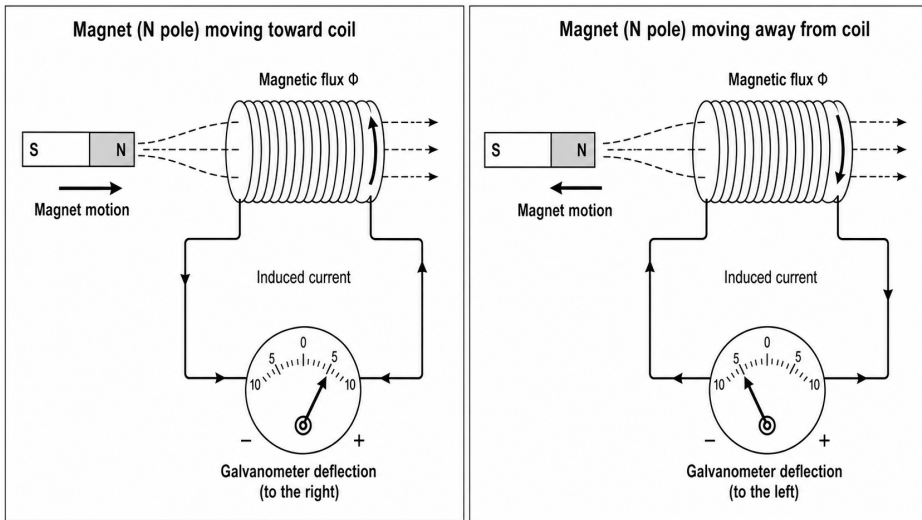


Figure 15: Figure 2.3: Induced current and galvanometer deflection when a bar magnet approaches and recedes from a coil.

of change of flux linkage.

For a coil of N turns in which each turn links flux Φ , the flux linkage is $N\Phi$ and the induced emf is

$$e = -N \frac{d\Phi}{dt} \quad (2.9)$$

where e is in volts. For many elementary problems the average induced emf is sufficient:

$$E_{avg} = \frac{N \Delta\Phi}{\Delta t} \quad (2.10)$$

Equation (2.10) gives magnitude; direction is fixed by Lenz's law.

Lenz's law

The negative sign in Equation (2.9) is explained by **Lenz's law**: the induced emf acts in a direction that opposes the change producing it. When the north pole of a magnet approaches a coil, the induced current establishes a flux that opposes the increase, so the

coil appears to resist the magnet's approach. When the magnet is withdrawn, the induced current reverses and tries to sustain the decaying flux. Lenz's law is a consequence of energy conservation: if the induced current supported rather than opposed the change, energy would appear without input work.

Dynamically induced emf

If a conductor physically moves through a magnetic field and cuts flux, the resulting emf is called **dynamically induced**. Consider a straight conductor of active length l moving with speed v perpendicular to a field of flux density B . In time Δt it sweeps an area $\Delta A = lv \Delta t$, cutting flux $\Delta \Phi = B lv \Delta t$. By Equation (2.10),

$$E = Blv \quad (2.11)$$

with B in tesla, l in metres, and v in metres per second. This is the basic emf relation behind every generator.

Worked Example 2.4

A straight conductor of length 0.25 m moves at 8 m/s at right angles to a field of flux density 0.6 T. Find the induced emf.

$$E = Blv = 0.6 \times 0.25 \times 8 = 1.2 \text{ V}$$

Statically induced emf

When a conductor does not move but the flux linking it changes in time, the resulting emf is **statically induced**. This occurs in a stationary coil when its own current, or the current in a nearby coil, changes. Two forms arise:

- **self-induced emf**, due to a coil's own changing current;

- **mutually induced emf**, due to a changing current in a neighbouring coil.

Self-inductance

When the current through a coil changes, the flux it produces changes, and that changing flux induces an emf in the same coil. This is **self-inductance**:

$$e = -L \frac{di}{dt} \quad (2.12)$$

where L is the self-inductance in henries (H). One henry corresponds to an induced emf of 1 V for a rate of current change of 1 A/s. By Lenz's law the sign is opposing: rising current induces an emf that retards it, and falling current induces an emf that tries to sustain it. This is why an inductor current cannot change instantaneously, and why breaking the current in a relay coil or motor winding produces a high-voltage spike across the contacts — an effect commonly suppressed by a freewheeling diode.

Worked Example 2.5

A coil of inductance 2 H carries a current that increases uniformly from 0 A to 3 A in 0.05 s. Find the magnitude of the induced emf.

$$|e| = L \frac{\Delta i}{\Delta t} = 2 \times \frac{3}{0.05} = 120 \text{ V}$$

The emf acts against the rise in current.

Mutual inductance

When two coils lie close enough that some of the flux produced by the first links the second, a change in the first coil's current induces an emf in the second. This is **mutual inductance**:

$$e_2 = -M \frac{di_1}{dt} \quad (2.13)$$

where M is the mutual inductance between the two coils. This is the principle of the transformer: changing current in the primary sets up a changing flux in the common core, which links the secondary and induces an emf there. No charge crosses from one winding to the other; energy transfers through the magnetic field alone, which is why transformer windings remain electrically isolated.

Flux linkage and coefficient of coupling

If a coil has N turns and each turn links the same flux Φ , the flux linkage is $N\Phi$. Inductance measures how strongly a change in current changes this flux linkage.

For two coils, not all the flux produced by one necessarily links the other. Some escapes as leakage flux, so the coupling between coils is rarely perfect. The closeness of coupling is described by the **coefficient of coupling**,

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

where L_1 and L_2 are the self-inductances and $0 \leq k \leq 1$. A value near 1 indicates tight coupling; a small value indicates loose coupling. Even in transformer windings on a common core, k is

slightly less than 1, which is why real transformers exhibit leakage reactance and voltage regulation.

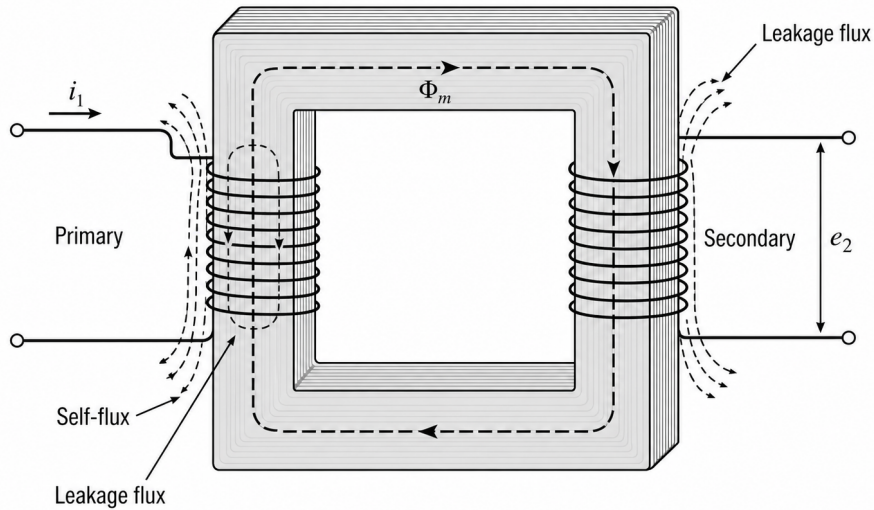


Figure 16: Figure 2.4: Mutual induction in two coils on a common core, showing mutual flux, self-flux, leakage flux, and induced secondary emf.

Worked Example 2.6

Two coils have mutual inductance 0.4 H. The current in the first coil decreases uniformly from 5 A to 1 A in 0.02 s. Find the magnitude of the induced emf in the second coil.

$$|e_2| = M \frac{|\Delta i_1|}{\Delta t} = 0.4 \times \frac{4}{0.02} = 80 \text{ V}$$

2.3 Direction Rules

The direction of an induced emf, or of the force on a current-carrying conductor, is set by the mutual orientation of field,

current, and motion. Two hand rules due to Fleming fix this direction at a glance: one for motor action, the other for generator action.

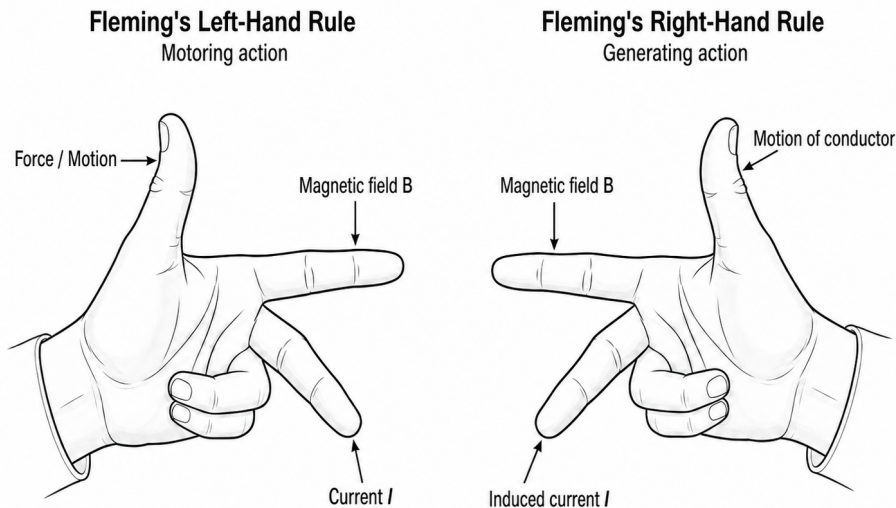


Figure 17: Figure 2.5: Fleming's left-hand rule for motoring action and right-hand rule for generating action.

Fleming's Left-Hand Rule

Fleming's Left-Hand Rule gives the direction of force on a current-carrying conductor in a magnetic field. Hold the thumb, forefinger, and middle finger of the left hand mutually perpendicular; then

- the **forefinger** points along the magnetic field,
- the **middle finger** points along the current, and
- the **thumb** points along the force or resulting motion.

This is the rule behind electric motors, moving-coil instruments, and loudspeakers.

Worked Example 2.7

A straight conductor carries current upward between the pole faces of a magnet, with the magnetic field directed from left to right in the viewer's frame. Find the direction of force on the conductor.

Applying Fleming's Left-Hand Rule with forefinger left-to-right and middle finger upward, the thumb points **out of the page, toward the observer**. The force is perpendicular to both the field and the current.

Fleming's Right-Hand Rule

Fleming's Right-Hand Rule gives the direction of induced current in a conductor moving through a magnetic field. Hold the thumb, forefinger, and middle finger of the right hand mutually perpendicular; then

- the **thumb** points along the motion of the conductor,
- the **forefinger** points along the magnetic field, and
- the **middle finger** points along the induced current.

This is the rule used for generators and for simple moving-conductor emf problems.

Worked Example 2.8

A conductor moves downward through a magnetic field directed from left to right in the viewer's frame. Find the direction of induced current.

Applying Fleming's Right-Hand Rule with thumb downward and forefinger left-to-right, the middle finger points **into the page**. The induced current in the conductor flows away from

the observer.

Table 2.2: Which direction rule to use

Situation	Physical effect	Rule to use
Current-carrying conductor in a magnetic field	Force or motion produced	Fleming's Left-Hand Rule
Conductor moving through a magnetic field	Induced emf or current produced	Fleming's Right-Hand Rule

Both rules give direction only; magnitudes come from $F = BIl$ or $E = Blv$. Conventional current direction is used throughout unless a problem states otherwise.

Worked Interpretation Exercise

Reading an Inductor Datasheet

Real components expose the ideas of this chapter in concise numerical form. Consider the specification table for the TDK PLEC69B thin-film power inductor, and in particular the part PLEC69BCA100M-1PT00:

- nominal inductance = $10\ \mu\text{H}$
- inductance tolerance = $\pm 20\%$
- inductance measuring frequency = $1\ \text{MHz}$
- maximum DC resistance = $1680\ \text{m}\Omega = 1.68\ \Omega$
- saturation-based current limit, $I_{\text{sat}} = 0.2\ \text{A}$

- temperature-rise-based current limit, $I_{temp} = 0.35 \text{ A}$
- operating temperature range = -40°C to $+125^\circ\text{C}$ including self-heating.

The datasheet defines the rated current as the smaller of I_{sat} and I_{temp} , so for this part

$$I_{rated} = 0.2 \text{ A}$$

The saturation limit, not the heating limit, sets the usable current: the core's inductance begins to collapse before the winding overheats. A $\pm 20\%$ tolerance on a $10 \mu\text{H}$ nominal value allows a measured inductance between $8 \mu\text{H}$ and $12 \mu\text{H}$ at the specified test frequency. The DC resistance sets the copper loss; at 0.1 A ,

$$P = I^2 R = (0.1)^2 \times 1.68 = 16.8 \text{ mW}$$

Small in absolute terms, but significant for a component this size. Each datasheet entry maps directly onto a chapter concept: inductance onto self-inductance, I_{sat} onto core saturation, DC resistance onto winding loss, and the temperature range onto safe operation.

Further Reading

1. A. E. Fitzgerald, Charles Kingsley Jr., and Stephen D. Umans, *Electric Machinery* — classic text for magnetic circuits, transformers, and electromechanical energy conversion.
2. Stephen J. Chapman, *Electric Machinery Fundamentals* — widely used machine text with accessible treatment of magnetic circuits, induction, transformers, and motors.
3. I. J. Nagrath and D. P. Kothari, *Electric Machines* — popular

text for magnetic circuits, electromagnetic induction, transformers, and rotating machines.

4. Edward Hughes, John Hiley, Keith Brown, and Ian McKenzie Smith, *Electrical and Electronic Technology* — broad introductory reference connecting magnetic effects with practical electrical equipment.

A.C. Circuits

{Overview

} This unit introduces alternating-current circuit analysis, including sinusoidal quantities, phase relationships, reactance, impedance, power in AC systems, and simple RLC behavior.

{Contents

}

Unit 3: A.C. Circuits 3.1 Basic AC Terms 3.2 Pure R, L, and C under Sinusoidal Excitation 3.3 Simple AC Circuits 3.4 AC Power 3.5 Three-Phase Connection Basics 3.6 Practice Set

Unit 3: A.C. Circuits

Most electrical systems around us do not operate on steady DC. The supply reaching homes, classrooms, workshops, and small industries is alternating current, and even equipment that runs internally on DC — battery chargers, computer power supplies, LED drivers — usually takes AC at its input. AC circuit theory is therefore central to electrical engineering.

This chapter develops the language and methods used for sinusoidal AC. It begins with waveform terminology and the RMS, average, form-factor, and peak-factor relations. It then examines the three basic circuit elements under sinusoidal excitation, combines them in series and parallel, and develops the phasor and complex-impedance methods needed for practical analysis. The chapter closes with AC power quantities and the voltage-current relations of balanced three-phase star and delta loads. It draws on Chapter 1 (passive components) and Chapter 2 (self-inductance) and prepares the ground for transformers, machines, measurements, and power systems.

3.1 Basic AC Terms

A torch powered by a battery carries current in a fixed direction; the 230 V, 50 Hz voltage at a wall outlet does not. It rises, falls to zero, reverses, reaches a negative peak, and returns again, repeat-

ing this pattern many times each second. This periodic reversal gives alternating current its name.

The **sinusoidal waveform** is central because it is produced naturally by alternators and because AC relations take simple, compact forms for a sine wave.

Instantaneous value and amplitude

The value of an AC quantity at a particular instant is its **instantaneous value**, written $v(t)$ or $i(t)$. For a sinusoidal voltage and current,

$$v(t) = V_m \sin \omega t \quad (3.1)$$

$$i(t) = I_m \sin \omega t \quad (3.2)$$

Here V_m and I_m are the **peak values** or **amplitudes**, measured from the zero axis, and ω is the angular speed in radians per second.

Cycle, time period, and frequency

A **cycle** is one complete repetition of the waveform. The **time period** T , in seconds, is the time taken for one cycle. The **frequency** f , in hertz, is the number of cycles per second:

$$f = \frac{1}{T} \quad (3.3)$$

A 50 Hz supply completes 50 cycles per second.

Angular speed

One sinusoidal cycle corresponds to 2π radians, so the **angular speed** or **angular frequency** is

$$\omega = 2\pi f \quad (3.4)$$

For 50 Hz, $\omega = 2\pi \times 50 \approx 314$ rad/s, a value that appears throughout power-frequency calculations.

Worked Example 3.1

An AC waveform has frequency 50 Hz. Find the time period and the angular speed.

$$T = \frac{1}{f} = \frac{1}{50} = 0.02 \text{ s} = 20 \text{ ms}$$

$$\omega = 2\pi f = 2\pi \times 50 \approx 314 \text{ rad/s}$$

RMS value

Because the instantaneous value of a sine wave varies continuously, comparison with DC — especially in terms of heating — requires a single equivalent figure. The **root mean square** or **RMS value** of an AC quantity is the DC value that would produce the same heating effect in a resistor. For a sinusoidal voltage and current,

$$V_{rms} = \frac{V_m}{\sqrt{2}}, \quad I_{rms} = \frac{I_m}{\sqrt{2}} \quad (3.5)$$

When the mains supply is called “230 V,” that value is the RMS. The corresponding peak is $V_m = \sqrt{2} V_{rms} \approx 325 \text{ V}$.

Average value

The average of a symmetrical sine wave over a full cycle is zero, since the positive and negative halves cancel. The practical figure of interest is therefore the **half-cycle average**:

$$V_{avg} = \frac{2V_m}{\pi} \approx 0.637 V_m, \quad I_{avg} = \frac{2I_m}{\pi} \approx 0.637 I_m \quad (3.6)$$

Form factor and peak factor

Two ratios characterise waveform shape:

$$\text{Form factor} = \frac{V_{rms}}{V_{avg}}, \quad \text{Peak factor} = \frac{V_m}{V_{rms}} \quad (3.7)$$

For a sine wave, the form factor is $0.707/0.637 \approx 1.11$ and the peak factor is $1/0.707 \approx 1.414$. Both figures are used in instrument interpretation, where a meter calibrated for a sine wave can be in error for non-sinusoidal inputs.

Worked Example 3.2

The RMS value of an AC supply is 230 V. Find the peak value, half-cycle average, form factor, and peak factor.

$$V_m = \sqrt{2} V_{rms} \approx 325.2 \text{ V}$$

$$V_{avg} = \frac{2V_m}{\pi} \approx 207.0 \text{ V}$$

$$\text{Form factor} = \frac{230}{207.0} \approx 1.11, \quad \text{Peak factor} = \frac{325.2}{230} \approx 1.414$$

Phase and phase difference

Two sine waves of the same frequency may not reach their peaks at the same instant. The relative angular position of a sinusoidal quantity is its **phase**, and the angular separation between two same-frequency waveforms is their **phase difference**. If they rise

and fall together they are **in phase**; if one peaks later it **lags**; if one peaks earlier it **leads**. A current that peaks one-quarter cycle after its voltage lags by 90° .

Phasors and complex-number representation

Drawing full sine waves every time two AC quantities are compared is cumbersome. Engineers represent a sinusoid of fixed frequency by a **phasor**: a vector whose length is the amplitude (or RMS value) and whose angle is the phase. In a circuit whose voltages and currents share one frequency, equal phasor directions mean in phase, a phasor drawn ahead by ϕ means it leads by ϕ , and one drawn behind means it lags. Phasor representation assumes a single common frequency; a diagram combining different-frequency components is not meaningful.

A phasor is represented algebraically by a complex number. In electrical engineering the imaginary unit is written j (because i denotes current), with

$$j = \sqrt{-1}, \quad j^2 = -1 \quad (3.8)$$

A complex number in rectangular form is $z = a + jb$; its magnitude and argument are

$$|z| = \sqrt{a^2 + b^2}, \quad \theta = \tan^{-1} \left(\frac{b}{a} \right) \quad (3.9)$$

The same quantity written in polar form is $z = |z| \angle \theta$. Rectangular form is convenient for addition and subtraction; polar form is convenient for multiplication, division, and reading magnitude and angle. A voltage phasor of magnitude V at angle θ is written

$$\tilde{V} = V\angle\theta = V(\cos\theta + j\sin\theta) \quad (3.10)$$

and the tilde distinguishes phasor quantities from time-domain instantaneous values.

A few conversions illustrate the arithmetic:

$$8 + j6 = 10\angle 36.9^\circ, \quad 12 - j5 = 13\angle(-22.6^\circ)$$

$$20\angle 30^\circ = 17.32 + j10, \quad 15\angle(-53.1^\circ) = 9 - j12$$

When converting from rectangular to polar, the quadrant of the point must be checked before accepting the value returned by $\tan^{-1}$.

Phase difference on an oscilloscope

On a CRO or DSO, two same-frequency sinusoids can be displayed on separate channels. Measuring the time shift Δt between corresponding points (such as positive-going zero crossings) and the time period T gives

$$\phi = \frac{\Delta t}{T} \times 360^\circ \quad (3.11)$$

A quarter-cycle shift corresponds to 90° . Modern digital scopes report phase directly in degrees, but the relation above remains the basis for interpreting the measurement.

Impedance

In a DC circuit, opposition to current is simply the resistance. In AC, resistors oppose current by resistance, inductors oppose changes in current by **inductive reactance**, and capacitors oppose

changes in voltage by **capacitive reactance**. The total opposition to AC is the **impedance** Z , measured in ohms. For the three ideal elements,

$$Z_R = R \quad (3.12)$$

$$X_L = \omega L = 2\pi fL, \quad Z_L = jX_L = j\omega L \quad (3.13)$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}, \quad Z_C = -jX_C = \frac{1}{j\omega C} \quad (3.14)$$

These forms are the foundation of AC circuit analysis.

Table 3.1: Basic AC waveform quantities for a sine wave

Quantity	Symbol	Relation
Time period	T	$1/f$
Angular speed	ω	$2\pi f$
RMS voltage	V_{rms}	$V_m/\sqrt{2}$
RMS current	I_{rms}	$I_m/\sqrt{2}$
Half-cycle average voltage	V_{avg}	$2V_m/\pi$
Half-cycle average current	I_{avg}	$2I_m/\pi$
Form factor	—	$V_{rms}/V_{avg} \approx 1.11$
Peak factor	—	$V_m/V_{rms} \approx 1.414$

3.2 Pure R, L, and C under Sinusoidal Excitation

The cleanest way to reach AC circuit behaviour is to study the three basic elements one at a time under the same sinusoidal excitation.

Pure resistance

Applying $v(t) = V_m \sin \omega t$ across a resistor gives, by Ohm's law,

$$i_R(t) = \frac{V_m}{R} \sin \omega t \quad (3.15)$$

Voltage and current share zero crossings and peaks. The phase difference is zero, and the impedance is $Z_R = R$. In RMS terms, $I_{rms} = V_{rms}/R$, so a purely resistive heater draws the same average power on AC as on DC at equal RMS voltage.

Pure inductance

For a sinusoidal voltage across a pure inductor,

$$i_L(t) = \frac{V_m}{X_L} \sin(\omega t - 90^\circ), \quad X_L = \omega L \quad (3.16)$$

The current lags the voltage by 90° : when the applied voltage is at its positive peak, the current is crossing zero; when the voltage reaches zero, the current is at its positive peak. The physical origin, developed in Chapter 2, is the self-induced emf that opposes any change in current. In RMS form, $I_{rms} = V_{rms}/X_L$.

Pure capacitance

For a sinusoidal voltage across a pure capacitor,

$$i_C(t) = \frac{V_m}{X_C} \sin(\omega t + 90^\circ), \quad X_C = \frac{1}{\omega C} \quad (3.17)$$

The current leads the voltage by 90° . A capacitor passes more current at higher frequency because rapid voltage changes move

charge onto and off its plates more quickly. In RMS form, $I_{rms} = V_{rms}/X_C$.

Figure 3.1 shows the reference circuits and the corresponding waveform and phasor relationships.

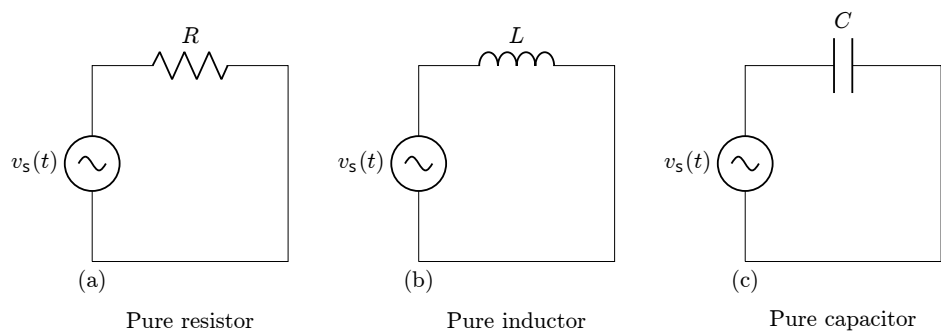


Figure 18: Figure 3.1a: Pure R, L, and C reference circuits

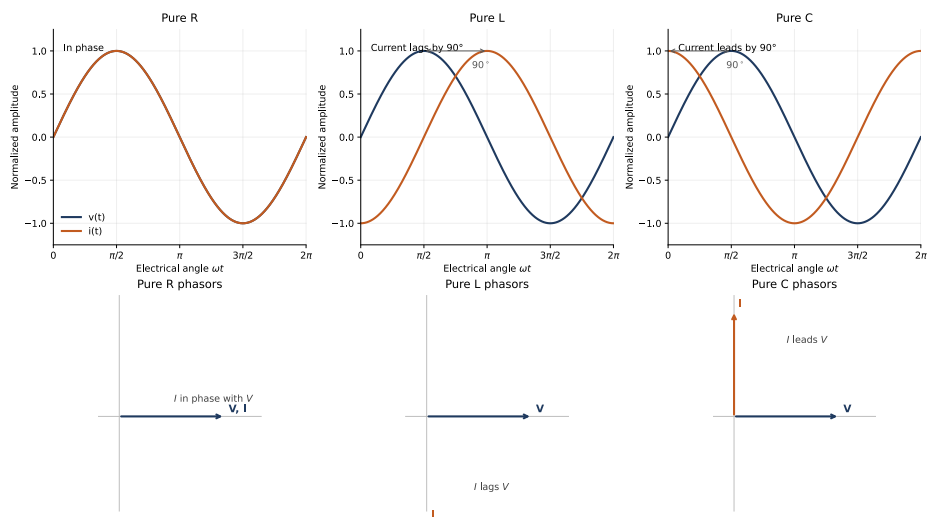


Figure 19: Figure 3.1b: Voltage-current waveforms and phasor relations for pure R, L, and C circuits

Frequency dependence of reactance

Inductive and capacitive reactance vary oppositely with frequency. Because $X_L = 2\pi fL$ increases with frequency while $X_C = 1/(2\pi fC)$ decreases with it, an inductor opposes high-frequency current more strongly and a capacitor opposes low-frequency current

more strongly. This contrast underlies filters, tuning, and many power-electronic circuits.

Worked Example 3.3

Find the inductive reactance of a 0.2 H coil at 50 Hz.

$$X_L = 2\pi fL = 2\pi \times 50 \times 0.2 \approx 62.8 \, \Omega$$

Worked Example 3.4

Find the capacitive reactance of a 20 μF capacitor at 50 Hz.

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 20 \times 10^{-6}} \approx 159.2 \, \Omega$$

Real components are never perfectly ideal. A practical resistor shows small parasitic inductance or capacitance at high frequency; a practical inductor has winding resistance, and its inductance can fall as the magnetic core approaches saturation. A practical capacitor has tolerance, leakage, a voltage rating, and frequency limits. Pure R, L, and C remain the right building blocks for ideal analysis, while equipment selection depends on the full set of component ratings.

3.3 Simple AC Circuits

Practical AC circuits combine resistance and reactance, so voltage and current are generally neither in phase nor 90° apart. Two facts carry most series and parallel calculations:

- In a **series** circuit, the same current flows through every element.

- In a **parallel** circuit, the same voltage appears across every branch.

R-L series circuit

With a common current through R and L , the resistor voltage V_R is in phase with the current and the inductor voltage V_L leads it by 90° . The source voltage is the phasor sum of V_R and V_L , so it leads the current by an angle between 0° and 90° — the circuit is lagging. The complex impedance, its magnitude, and phase angle are

$$Z_{RL} = R + jX_L \quad (3.18)$$

$$|Z_{RL}| = \sqrt{R^2 + X_L^2}, \quad \tan \phi = \frac{X_L}{R} \quad (3.19)$$

and the current follows from the AC form of Ohm's law,

$$\tilde{I} = \frac{\tilde{V}}{\tilde{Z}} \quad (3.20)$$

Worked Example 3.5

An R-L series circuit with $R = 30 \, \Omega$ and $X_L = 40 \, \Omega$ is connected to 230 V AC. Find the impedance, current, and phase angle.

Taking magnitudes first,

$$Z = \sqrt{30^2 + 40^2} = 50 \, \Omega, \quad I = \frac{230}{50} = 4.6 \, \text{A}$$

$$\tan \phi = \frac{40}{30} = 1.333, \quad \phi \approx 53.1^\circ$$

The same result in complex form, with the supply voltage as

reference, is

$$\tilde{Z} = 30 + j40 = 50\angle 53.1^\circ \Omega$$

$$\tilde{I} = \frac{230\angle 0^\circ}{50\angle 53.1^\circ} = 4.6\angle(-53.1^\circ) \text{ A}$$

The negative angle confirms that the current lags the supply by 53.1° . Subsequent examples use whichever form is more direct.

R-C series circuit

The resistor voltage remains in phase with the current while the capacitor voltage lags the current by 90° , so the source voltage lags the current and the circuit is leading. The impedance forms are

$$Z_{RC} = R - jX_C \quad (3.21)$$

$$|Z_{RC}| = \sqrt{R^2 + X_C^2}, \quad \tan \phi = \frac{X_C}{R} \quad (3.22)$$

Worked Example 3.6

An R-C series circuit with $R = 40 \Omega$ and $X_C = 30 \Omega$ is connected to 200 V AC. Find the current phasor.

$$\tilde{Z} = 40 - j30 = 50\angle(-36.9^\circ) \Omega$$

$$\tilde{I} = \frac{200\angle 0^\circ}{50\angle(-36.9^\circ)} = 4\angle 36.9^\circ \text{ A} = 3.2 + j2.4 \text{ A}$$

The current leads the supply voltage by 36.9° .

R-L-C series circuit

When a resistor, inductor, and capacitor are in series, the inductive and capacitive reactances act oppositely, giving a net reactance $X = X_L - X_C$. The impedance and phase angle are therefore

$$Z_{RLC} = R + j(X_L - X_C) \quad (3.23)$$

$$|Z_{RLC}| = \sqrt{R^2 + (X_L - X_C)^2}, \quad \tan \phi = \frac{X_L - X_C}{R} \quad (3.24)$$

If $X_L > X_C$ the circuit is net inductive and the current lags; if $X_C > X_L$ the circuit is net capacitive and the current leads; if $X_L = X_C$ the impedance is purely resistive at that frequency and the current is in phase with the voltage. A detailed treatment of the last case (resonance) is left to a later course.

Figure 3.2 shows waveform and phasor interpretations for the three series circuits, including the three phase outcomes of the R-L-C case.

Because the same current flows through every element in a series circuit,

$$V_R = IR, \quad V_L = IX_L, \quad V_C = IX_C \quad (3.25)$$

The supply voltage, however, is not the arithmetic sum $V_R + V_L + V_C$; the voltages are not in phase and must be combined phasorially. For a series R-L-C circuit,

$$V = \sqrt{V_R^2 + (V_L - V_C)^2} \quad (3.26)$$

In reactive circuits, V_L or V_C can individually exceed the supply voltage.

Worked Example 3.7

An R-L-C series circuit with $R = 20 \, \Omega$, $X_L = 50 \, \Omega$, $X_C = 30 \, \Omega$ is connected to 230 V, 50 Hz. Find the current, phase angle, and voltage across each element.

Net reactance and impedance:

$$X = X_L - X_C = 20 \, \Omega, \quad Z = \sqrt{20^2 + 20^2} \approx 28.3 \, \Omega$$

Current and phase angle:

$$I = \frac{230}{28.3} \approx 8.13 \, \text{A}, \quad \tan \phi = 1, \quad \phi = 45^\circ \text{ (lagging)}$$

Element voltages:

$$V_R = 8.13 \times 20 \approx 162.6 \, \text{V}$$

$$V_L = 8.13 \times 50 \approx 406.5 \, \text{V}, \quad V_C = 8.13 \times 30 \approx 243.9 \, \text{V}$$

Although V_L and V_C individually exceed the supply, the phasor combination is consistent:

$$V = \sqrt{162.6^2 + (406.5 - 243.9)^2} \approx 230 \, \text{V}$$

Parallel circuits and admittance

In a parallel AC circuit the supply voltage is common to every branch while the branch currents generally differ in magnitude and phase. For the three basic branches fed from a voltage V ,

$$I_R = \frac{V}{R}, \quad I_L = \frac{V}{X_L}, \quad I_C = \frac{V}{X_C} \quad (3.27)$$

with I_R in phase with V , I_L lagging by 90° , and I_C leading by 90° . The total line current is the phasor sum. The treatment is therefore the mirror image of the series case: in series circuits, current is common and voltages are added phasorially; in parallel circuits, voltage is common and currents are added phasorially.

The reciprocal of impedance is the **admittance**:

$$Y = \frac{1}{Z} = G + jB \quad (3.28)$$

where G is the **conductance** and B is the **susceptance**, both in siemens. For the three pure elements,

$$Y_R = \frac{1}{R}, \quad Y_L = -\frac{j}{X_L}, \quad Y_C = \frac{j}{X_C} \quad (3.29)$$

The sign pattern reflects the phase relations: in an inductive branch the current lags the voltage, so the imaginary part of Y is negative; in a capacitive branch the current leads, so the imaginary part is positive. Impedances add directly in series; admittances add directly in parallel. For a parallel R-L-C circuit the total admittance is therefore

$$Y_{RLC} = \frac{1}{R} + j\left(\frac{1}{X_C} - \frac{1}{X_L}\right) \quad (3.30)$$

with magnitude $|Y| = \sqrt{G^2 + B^2}$, supply current $\tilde{I} = \tilde{V}\tilde{Y}$, and total impedance $Z = 1/Y$. The magnitude of the line current can also be written

$$I = \sqrt{I_R^2 + (I_L - I_C)^2} \quad (3.31)$$

The branch-current method finds each branch current and then combines the currents phasorially. The admittance method is more compact when the number of branches grows.

Worked Example 3.8

A parallel R-L-C circuit with $R = 100 \, \Omega$, $L = 0.2 \, \text{H}$, $C = 20 \, \mu\text{F}$ is connected to 230 V, 50 Hz. Find the branch currents, the total current, and the total impedance.

Reactances:

$$X_L = 2\pi \times 50 \times 0.2 \approx 62.8 \, \Omega, \quad X_C \approx 159.2 \, \Omega$$

Branch currents:

$$I_R = \frac{230}{100} = 2.3 \, \text{A}, \quad I_L = \frac{230}{62.8} \approx 3.66 \, \text{A}, \quad I_C = \frac{230}{159.2} \approx 1.44 \, \text{A}$$

In phasor form, with the supply voltage as reference,

$$\tilde{I} = 2.3 - j3.66 + j1.44 = 2.3 - j2.22 \, \text{A}$$

$$|\tilde{I}| \approx 3.20 \, \text{A}, \quad \phi \approx -44.0^\circ$$

so $\tilde{I} \approx 3.20 \angle (-44.0^\circ) \, \text{A}$. The inductive current dominates

and the circuit is net inductive. The total admittance and impedance follow directly:

$$\tilde{Y} = \frac{\tilde{I}}{\tilde{V}} \approx 0.0139 \angle (-44.0^\circ) \text{ S}, \quad \tilde{Z} = \frac{1}{\tilde{Y}} \approx 71.9 \angle 44.0^\circ \Omega$$

Problem-solving method

For a consistent approach to mixed AC circuits:

1. List R , L , C , frequency, and supply voltage. Compute X_L and X_C if needed.
2. For a series circuit, form the total impedance and divide the supply voltage by it; then find element voltages from the common current. For a parallel circuit, compute each branch current from the common voltage, combine them phasorially, and obtain total impedance from $Z = V/I$ or $Z = 1/Y$.
3. Identify the circuit as lagging, leading, or in phase from the sign of the net reactance or net susceptance.
4. Compute power factor or power quantities if the problem calls for them.

Careful distinction between the common quantity (current in series, voltage in parallel) and the phasor-added quantity (voltage in series, current in parallel) removes most routine errors.

Table 3.2: Quick comparison of common AC circuits

Circuit	Impedance	Current	
		relation	Nature
Pure R	$Z = R$	I in phase with V	purely resistive
Pure L	$Z = X_L$	I lags V by 90°	inductive
Pure C	$Z = X_C$	I leads V by 90°	capacitive
Series RL	$\sqrt{R^2 + X_L^2}$	current lags	lagging
Series RC	$\sqrt{R^2 + X_C^2}$	current leads	leading
Series RLC	$\sqrt{R^2 + (X_L - X_C)^2}$	depends on X_L vs X_C	mixed
Parallel RL	use admittance	total current lags	lagging
Parallel RC	use admittance	total current leads	leading
Parallel RLC	use admittance	depends on I_L vs I_C	mixed

3.4 AC Power

Power in an AC circuit needs more care than in a DC circuit. In DC, $P = VI$ gives the power directly. In AC, voltage and current are generally not in phase, so not all of the product VI represents useful conversion into heat, light, or mechanical work.

Instantaneous power in R, L, and C

The product $p(t) = v(t)i(t)$ is the **instantaneous power**. In a pure resistor, voltage and current are in phase, so $p(t)$ is always non-negative; the resistor continuously absorbs energy and dissipates it as heat. In a pure inductor, current lags voltage by 90° and $p(t)$ alternates equally between positive and negative values: during the positive intervals, energy is stored in the magnetic field; during the negative intervals, the stored energy is returned to the source. A pure capacitor behaves analogously, storing and returning energy through the electric field. Over a full cycle, the average power in an ideal inductor or capacitor is zero, while the average power in a resistor is not.

Active, reactive, and apparent power

The three standard power quantities are

$$P = V_{rms} I_{rms} \cos \phi \quad (3.32)$$

$$Q = V_{rms} I_{rms} \sin \phi \quad (3.33)$$

$$S = V_{rms} I_{rms} \quad (3.34)$$

P is the **active** (or true) power in watts, representing useful conversion. Q is the **reactive power** in volt-ampere reactive (var), representing energy exchanged with reactive elements. S is the **apparent power** in volt-ampere (VA). They are related by

$$S^2 = P^2 + Q^2 \quad (3.35)$$

For a pure resistor, $\phi = 0$, so $P = S$ and $Q = 0$. For a pure inductor or capacitor, $\phi = \pm 90^\circ$ and the average active power is zero.

Power factor, impedance triangle, and power triangle

The ratio of active to apparent power is the **power factor**:

$$\cos \phi = \frac{P}{S} \quad (3.36)$$

In a series AC circuit it also equals R/Z . A lagging current gives a lagging power factor, a leading current a leading power factor. Low power factor is costly because it forces a larger line current for the same useful power, increasing conductor losses and voltage drop. Two right triangles summarise the geometry. The **impedance triangle** has R as its base, net reactance ($X_L - X_C$) as its perpendicular, and Z as its hypotenuse, with angle ϕ at the base. The **power triangle** has P as its base, Q as its perpendicular, and S as its hypotenuse, with the same angle ϕ . Accordingly,

$$\cos \phi = \frac{P}{S}, \quad \sin \phi = \frac{Q}{S}, \quad \tan \phi = \frac{Q}{P} \quad (3.37)$$

Figure 3.3 shows both triangles side by side.

Worked Example 3.9

A single-phase load draws 8 A at 230 V with a lagging power factor of 0.8. Find the apparent, active, and reactive powers.

$$S = 230 \times 8 = 1840 \text{ VA}$$

$$P = 1840 \times 0.8 = 1472 \text{ W}$$

With $\sin \phi = \sqrt{1 - 0.8^2} = 0.6$,

$$Q = 1840 \times 0.6 = 1104 \text{ var}$$

Many workshop and industrial loads, especially motors and transformers with inductive windings, operate at a lagging power factor. Power-factor correction — typically by shunt capacitors — is consequently a standard topic in electrical installations.

3.5 Three-Phase Connection Basics

Industrial systems use **three-phase AC** because it is efficient for power transmission, distribution, and motor operation. In a balanced three-phase system, three sinusoidal phase voltages of equal magnitude and common frequency are displaced by 120° from each other. The focus here is the voltage-current relations for star and delta connections; generation details belong to later courses.

Figure 3.4 shows balanced star and delta loads side by side.

Star connection

In a **star** (or wye) connection, one end of each phase winding meets at a common neutral point, and the other ends go to the three line conductors. For a balanced star-connected load,

$$V_L = \sqrt{3} V_{ph}, \quad I_L = I_{ph} \quad (3.38)$$

A 400 V line-to-line system therefore has a phase voltage of $400/\sqrt{3} \approx 231$ V, which matches the 230 V class of single-phase

loads.

Delta connection

In a **delta** connection, the three phases form a closed loop and the line conductors are tapped from the three vertices. For a balanced delta-connected load,

$$V_L = V_{ph}, \quad I_L = \sqrt{3} I_{ph} \quad (3.39)$$

So in star the voltage scales by $\sqrt{3}$ between line and phase while the current does not; in delta the current scales by $\sqrt{3}$ while the voltage does not.

Worked Example 3.10

A balanced star-connected load is fed at 400 V line voltage with a 10 A line current. Find the phase voltage and phase current.

$$V_{ph} = \frac{400}{\sqrt{3}} \approx 231 \text{ V}, \quad I_{ph} = I_L = 10 \text{ A}$$

Worked Example 3.11

A balanced delta-connected load is fed at 400 V line voltage with a 17.32 A line current. Find the phase voltage and phase current.

$$V_{ph} = V_L = 400 \text{ V}, \quad I_{ph} = \frac{17.32}{\sqrt{3}} = 10 \text{ A}$$

Table 3.3: Star and delta relations for balanced loads

Connection	Line vs phase voltage	Line vs phase current
Star (Y)	$V_L = \sqrt{3} V_{ph}$	$I_L = I_{ph}$
Delta (Δ)	$V_L = V_{ph}$	$I_L = \sqrt{3} I_{ph}$

3.6 Practice Set

Direct formula questions

1. A sinusoidal supply has frequency 50 Hz. Find its time period.
2. The RMS voltage of a sine wave is 230 V. Find its peak value.
3. Find the inductive reactance of a 0.1 H inductor at 50 Hz.
4. Find the capacitive reactance of a 50 μF capacitor at 50 Hz.

Phasor and graph questions

1. In a pure inductive circuit with voltage as reference, where is the current phasor drawn?
2. In a pure capacitive circuit, how does the current waveform shift compare with the voltage wave?
3. In an R-L series circuit, why is the supply voltage not the arithmetic sum of V_R and V_L ?
4. What does the instantaneous-power waveform of a pure capacitor imply about its average power?

Complex-impedance questions

1. Convert $6 + j8$ to polar form.
2. Convert $20\angle 60^\circ$ to rectangular form.
3. A series circuit has $R = 25 \Omega$ and $X_L = 15 \Omega$. Write its impedance in rectangular and polar form.

4. A series circuit has $R = 12\ \Omega$, $X_L = 20\ \Omega$, $X_C = 8\ \Omega$. Find the total impedance and state whether the current leads or lags.

Admittance and parallel-circuit questions

1. A $100\ \Omega$ resistor is connected across 200 V AC. Find the branch current.
2. A parallel combination of $R = 100\ \Omega$ and $X_C = 50\ \Omega$ is connected across 200 V AC. Find the branch currents and state whether the total current leads or lags.
3. Write the admittance of a pure inductive branch in complex form and explain the sign of its imaginary part.
4. A parallel R-L-C circuit has $G = 0.02\ \text{S}$ and $B = -0.015\ \text{S}$. Find $|Y|$ and state whether the circuit is net inductive or net capacitive.

Worked Interpretation Exercise

Reading AC Instrument and Component Specifications

AC quantities are often read through an instrument specification. A representative true-RMS digital multimeter may have AC-voltage RMS ranges of 6.000 V, 60.00 V, and 600.0 V; AC-voltage accuracy of 1.0% + 3 from 45 Hz to 500 Hz; frequency measurement to 50.00 kHz; and a CAT III 600 V safety rating. The True-RMS designation means the meter reports RMS voltage directly, so a display of 230.0 V on a wall socket is $230\ \text{V}_{rms}$, not a peak value. The 600 V top range accommodates both 230 V single-phase and 400 V three-phase line-to-line measurements, though safe use depends on correct leads and technique. The specified accuracy applies from 45 Hz to 500 Hz, comfortably spanning 50 Hz

mains. The frequency counter extends to 50 kHz, which is useful for signal and drive-frequency work even though it is not needed at power frequency. The CAT III 600 V rating is a safety-category statement under IEC 61010, not a guarantee that any 600 V measurement is safe.

A motor-run capacitor illustrates the component side. A capacitor with ordering code B32333I6305J070 is specified as 3 μF with $\pm 5\%$ tolerance, a 450 V AC rating, an IEC 60252-1 life expectancy of 30 000 h, and safety class S2. The tolerance places the actual capacitance between 2.85 μF and 3.15 μF . The AC voltage rating confirms that the part is designed for continuous AC service in single-phase motors such as ceiling fans and pumps, not merely for DC filtering. Selection therefore depends not only on the reactance needed but also on voltage rating, operating life, and safety class.

Further Reading

1. Charles K. Alexander and Matthew N. O. Sadiku, *Fundamentals of Electric Circuits* — widely used reference for sinusoidal steady-state analysis, phasors, impedance, and AC power.
2. William H. Hayt, Jack E. Kemmerly, and Steven M. Durbin, *Engineering Circuit Analysis* — standard circuit-analysis text with rigorous AC circuit and phasor treatment.
3. James W. Nilsson and Susan A. Riedel, *Electric Circuits* — popular text for AC steady-state methods, complex impedance, and power calculations.
4. Edward Hughes, John Hiley, Keith Brown, and Ian McKenzie Smith, *Electrical and Electronic Technology* — practical intro-

duction to AC quantities, three-phase systems, and electrical equipment.

Transformer and Machines

{Overview

} This unit presents the basic construction, principles, and operating ideas of transformers and electrical machines, with emphasis on conceptual understanding before detailed machine theory.

{Contents

}

Unit 4: Transformer and Machines 4.1 Single-Phase Transformer 4.2 AC and DC Motors

Unit 4: Transformer and Machines

Electrical energy becomes truly useful when it can be changed from one voltage level to another and when it can be converted into motion. A **transformer** changes AC voltage and current levels without changing frequency. An **electric motor** converts electrical energy into mechanical rotation. These two ideas appear everywhere: in distribution transformers on electric poles, in mobile-phone chargers, in welding sets, in ceiling fans, in water pumps, in workshop machines, and in industrial drives.

This chapter grows naturally from Chapters 2 and 3. Chapter 2 explained electromagnetic induction and magnetic circuits. Chapter 3 explained AC quantities, reactance, and power. Here those ideas become physical devices. We first study the single-phase transformer: its main parts, working principle, emf equation, transformation ratio, ratings, and simple calculations. Then we study AC and DC motors through their parts, principles, simple performance ideas, and practical applications.

Later chapters on power supplies, semiconductor rectifiers, instrumentation, and industrial control will rely on the equipment introduced here. A charger uses a transformer. A pump uses a motor. A control panel may contain both.

4.1 Single-Phase Transformer

Suppose a workshop needs a low-voltage AC supply for a control circuit, a soldering tool, or a battery charger front end. The building supply may be 230 V, 50 Hz, but the equipment may need 12 V, 24 V, or some other value. A transformer makes that change possible. In a power system, the same principle is used in the opposite direction: voltage is stepped up for transmission and stepped down for utilization.

The basic transformer has no rotating parts. That makes it very different from a motor. It transfers energy from one circuit to another through a changing magnetic field. The primary and secondary windings are electrically isolated, but magnetically linked.

Image prompt for Figure 4.1: Create a clean textbook-style illustration of a single-phase two-winding transformer on a laminated core. Show the primary winding on one limb connected to an AC source labeled V_1 , the secondary winding on the other limb connected to a load labeled V_2 , and alternating core flux Φ linking both windings. Label laminated core, primary winding, secondary winding, input current I_1 , output current I_2 , and load.

Main parts of a two-winding transformer

A two-winding transformer has these main physical parts:

- **Magnetic core:** usually built from thin laminated silicon-steel sheets for power-frequency transformers, so that magnetic reluctance is low and eddy-current loss is reduced.
- **Primary winding:** the winding connected to the supply.
- **Secondary winding:** the winding connected to the load.
- **Insulation:** separates turns of the same winding and also sep-

arates primary from secondary for safety.

- **Tank or enclosure:** present in many practical transformers for protection and cooling.
- **Terminals or leads:** provide electrical connection to the outside circuit.

In small transformers, the windings may be arranged one over the other on the same limb. In larger transformers, the construction becomes more elaborate, but the teaching idea remains the same: one winding creates alternating flux in the core, and that changing flux induces emf in the other winding.

Core-type and shell-type idea

Textbooks often distinguish between **core-type** and **shell-type** transformers. This distinction is useful because students will see both forms in books and laboratories.

In a **core-type transformer**, the windings surround a considerable part of the core. In a **shell-type transformer**, the core surrounds a larger part of the windings. Both constructions provide:

- a magnetic path of low reluctance,
- magnetic linking between primary and secondary,
- space for insulated windings,
- and a practical structure for cooling and mounting.

Basic working principle

The transformer works on **mutual induction**. When AC voltage is applied to the primary winding, an alternating current flows in the primary. That alternating current produces an alternating magnetic flux in the core. Because the core flux links the secondary winding also, an emf is induced in the secondary.

This is exactly the Chapter 2 idea of Faraday's law applied to two coils on a common magnetic path. If the flux changes with time, emf is induced. If the flux is steady, no transformer action occurs. That is why a transformer requires a changing magnetic field and therefore normally works with AC rather than steady DC.

The physical sequence is:

1. AC supply is applied to the primary.
2. Primary current creates alternating core flux.
3. Core flux links both windings.
4. Induced emf appears in both primary and secondary.
5. If a load is connected, secondary current flows and power is delivered to the load.

Why DC does not work in an ordinary transformer

Students often ask this question because the primary is just a coil on a core. The answer follows directly from Faraday's law. If the applied voltage is pure DC, the current becomes steady after the switching transient. Then the magnetic flux also becomes nearly steady. A steady flux does not induce continuous emf in the secondary.

There is another practical danger. The winding resistance of a transformer is usually small. If DC is applied directly, the core does not present the usual AC magnetizing behavior, so the current may become excessively large. This can overheat the winding quickly. So a normal power transformer is intended for AC of its rated frequency, not direct connection to DC.

Induced emf and the transformer emf equation

We now derive the most important formula of this chapter. Assume the core flux is sinusoidal:

$$\Phi = \Phi_m \sin \omega t$$

where Φ_m is the maximum flux in weber and $\omega = 2\pi f$ is the angular frequency.

For a winding of N turns, Faraday's law gives the instantaneous induced emf:

$$e = -N \frac{d\Phi}{dt}$$

Differentiate the flux:

$$\frac{d\Phi}{dt} = \omega \Phi_m \cos \omega t$$

So the maximum value of induced emf is

$$E_m = \omega N \Phi_m = 2\pi f N \Phi_m$$

For a sinusoidal wave, RMS value is peak value divided by $\sqrt{2}$. Therefore,

$$E = \frac{E_m}{\sqrt{2}} = \frac{2\pi f N \Phi_m}{\sqrt{2}}$$

This becomes

$$\boxed{E = 4.44 f N \Phi_m} \quad (4.1)$$

Equation (4.1) is the **transformer emf equation**. It says that induced RMS emf depends on four things:

- frequency f ,
- number of turns N ,
- maximum flux Φ_m ,
- and the sinusoidal nature of the waveform, which gives the factor 4.44.

Applying Equation (4.1) to the primary and secondary windings separately:

$$\boxed{E_1 = 4.44fN_1\Phi_m} \quad (4.2)$$

$$\boxed{E_2 = 4.44fN_2\Phi_m} \quad (4.3)$$

Here:

- E_1 is primary induced emf,
- E_2 is secondary induced emf,
- N_1 is number of primary turns,
- N_2 is number of secondary turns.

Dividing Equation (4.3) by Equation (4.2):

$$\boxed{\frac{E_2}{E_1} = \frac{N_2}{N_1}} \quad (4.4)$$

This is the voltage-turns relation of the transformer.

Transformation ratio

The ratio of secondary quantity to primary quantity is often called the **transformation ratio**. In simple voltage form,

$$\boxed{k = \frac{V_2}{V_1} \approx \frac{E_2}{E_1} = \frac{N_2}{N_1}} \quad (4.5)$$

For an ideal or nearly ideal transformer, induced emf ratio, voltage ratio, and turns ratio are commonly treated as the same ratio.

Three important cases follow:

- If $N_2 > N_1$, then $V_2 > V_1$: this is a **step-up transformer**.
- If $N_2 < N_1$, then $V_2 < V_1$: this is a **step-down transformer**.
- If $N_2 = N_1$, then $V_2 \approx V_1$: this is an **isolation transformer**.

Transformers change voltage and current level, but not frequency.

A 50 Hz supply remains 50 Hz at the output.

Voltage ratio, current ratio, and power idea

If the transformer is ideal, input power equals output power:

$$V_1 I_1 = V_2 I_2 \quad (4.6)$$

From this,

$$\boxed{\frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{N_1}{N_2}} \quad (4.7)$$

If voltage is stepped down, current capability increases. If voltage is stepped up, current capability decreases.

That is why a small secondary voltage winding may use thicker wire. Lower voltage often means higher current for the same VA rating.

No-load idea and load idea

When the secondary is open and no load is connected, the transformer is said to be on **no load**. In that case:

- the primary still draws a small current from the supply,
- that current is needed mainly to establish alternating flux in the core,

- and to supply core losses.

When a load is connected to the secondary, secondary current flows. That current produces its own magnetic effect. The primary then draws additional current from the supply so that the core flux remains approximately at the required value and the transformer can transfer power to the load.

The transformer is not “sending current across the core.” Instead, the supply, core flux, and both windings interact magnetically so that energy is transferred from input to output.

Practical transformer losses

No practical transformer is perfect. Two important losses are:

- **Core loss:** mainly hysteresis loss and eddy-current loss in the core. These were introduced in Chapter 2.
- **Copper loss:** I^2R loss in the primary and secondary windings.

At no load, core loss is more prominent than copper loss because current is small. As load increases, copper loss increases because winding current increases.

This is why a transformer may become warm even when it is not heavily loaded. The core is still undergoing repeated magnetization and demagnetization at supply frequency.

Ratings found on practical transformers

A practical transformer is usually marked by:

- primary voltage,
- secondary voltage or voltages,
- frequency,
- VA or kVA rating,
- current rating,

- sometimes center tap or multiple taps,
- insulation and temperature information.

The rating is commonly given in VA rather than watts because transformer heating depends strongly on voltage and current, while the load power factor may vary.

For example, a 230 V / 12 V, 24 VA transformer can ideally supply about

$$I_2 = \frac{24}{12} = 2 \text{ A}$$

at the rated secondary voltage.

Center-tapped secondary

Some transformers have a **center-tapped** secondary. If a secondary is marked 12.6 V C.T., it means there are two equal halves with a center connection. The voltage from one end to the other full winding is 12.6 V. The voltage from the center tap to either end is half of that, ideally 6.3 V.

This is useful in:

- full-wave rectifier circuits,
- dual-polarity supply arrangements,
- heating and low-voltage legacy circuits,
- some charger and control applications.

Students should not confuse a center-tapped winding with two fully independent secondary windings. The winding is continuous, with a connection brought out from the midpoint.

Worked Example 4.1

A single-phase transformer has 500 turns on the primary and 100 turns on the secondary. The primary is connected to 230 V, 50 Hz AC. Assuming an ideal transformer, find the secondary voltage.

Use Equation (4.5):

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

Substitute the values:

$$\frac{V_2}{230} = \frac{100}{500} = 0.2$$

So

$$V_2 = 230 \times 0.2 = 46 \text{ V}$$

So the transformer is a step-down transformer and the secondary voltage is 46 V.

Worked Example 4.2

A transformer operates at 50 Hz. The primary winding has 400 turns and the maximum core flux is 2.5×10^{-3} Wb. Find the induced primary emf.

Use Equation (4.2):

$$E_1 = 4.44 f N_1 \Phi_m$$

Substitute:

$$E_1 = 4.44 \times 50 \times 400 \times 2.5 \times 10^{-3}$$

First multiply step by step:

$$50 \times 400 = 20000$$

$$20000 \times 2.5 \times 10^{-3} = 50$$

Now:

$$E_1 = 4.44 \times 50 = 222 \text{ V}$$

So the induced primary emf is 222 V.

This value is close to the kind of voltage seen in practical 230 V class transformers, which helps the formula feel physically reasonable.

Worked Example 4.3

An ideal transformer steps down 230 V to 23 V. If the secondary current is 4 A, find the primary current.

Using ideal power balance,

$$V_1 I_1 = V_2 I_2$$

So

$$230 I_1 = 23 \times 4 = 92$$

Therefore,

$$I_1 = \frac{92}{230} = 0.4 \text{ A}$$

So the primary current is **0.4 A**.

This result shows the inverse relation clearly. The voltage is reduced by a factor of 10, so the current is increased by a factor of 10.

4.2 AC and DC Motors

Why motors matter

A motor is an energy-conversion device. It takes electrical energy and produces mechanical rotation. That rotation may drive a ceiling fan, a water pump, a drilling machine, a conveyor, a wiper, or a printing mechanism.

This section focuses on the main motor ideas:

- what the main parts are,
- how torque is produced,
- how AC and DC motor arrangements differ,
- and where each type is commonly used.

The aim is a clear physical picture of how motors are built, how they produce torque, and how they are selected for common applications.

Image prompt for Figure 4.2: Create a clean textbook-style comparison diagram showing a basic DC motor and a basic AC induction motor side by side in cutaway form. For the DC motor, label yoke, poles, field winding or permanent magnet, armature, commutator, brushes, shaft, and bearings. For the AC induction mo-

tor, label stator, stator winding, squirrel-cage rotor, shaft, bearings, cooling fan, frame, and terminal box.

Common parts of rotating electrical machines

Many motors, whether AC or DC, share some broad mechanical parts:

- **Stator:** stationary part that carries windings or magnetic poles.
- **Rotor:** rotating part mounted on the shaft.
- **Shaft:** transmits mechanical power to the load.
- **Bearings:** support the shaft and allow smooth rotation.
- **Frame or yoke:** provides mechanical support and protection.
- **Fan or cooling arrangement:** removes heat in many practical machines.
- **Terminal box or leads:** used for electrical connection.

The electrical details differ from one motor type to another, but this common vocabulary helps identify machines in the lab and field.

Basic force and torque idea

The essential motor principle was introduced in Chapter 2 through Fleming's Left-Hand Rule. When a current-carrying conductor is placed in a magnetic field, it experiences a force. If conductors are arranged on a rotating armature or rotor, the forces combine to produce **torque**, which causes rotation.

So a motor needs two things:

- magnetic field,
- and current in conductors arranged so that force produces turning effect.

The difference between motor types lies in how the field is pro-

duced, how current reaches the rotating part, and how continuous unidirectional torque is maintained.

DC motor: main parts

A basic DC motor contains:

- **yoke or frame**,
- **field poles** and field winding, or permanent magnets,
- **armature** on the rotor,
- **commutator**,
- **brushes**,
- **shaft and bearings**.

The field system provides the magnetic field. The armature carries current-carrying conductors. The commutator and brushes are the distinctive parts of a classical brushed DC motor.

Working principle of a DC motor

When DC is supplied:

1. the field system produces magnetic flux,
2. current flows in armature conductors,
3. each conductor experiences force in the magnetic field,
4. the forces combine to produce torque,
5. the armature rotates.

If current in the rotating armature were not reversed at the correct instants, the torque would reverse every half turn and continuous rotation would not be maintained. The **commutator** solves this problem by reversing current in the armature connections as the rotor turns. The **brushes** provide sliding electrical contact with the commutator segments.

This is why the commutator is often called a mechanical rectifying

arrangement for the armature current direction.

Back emf in a DC motor

As the armature rotates in the magnetic field, it also behaves like a generator. An emf is induced in the armature conductors. This induced emf opposes the supply voltage, so it is called **back emf**.

The armature circuit is written as

$$\boxed{V = E_b + I_a R_a} \quad (4.8)$$

where:

- V is supply voltage,
- E_b is back emf,
- I_a is armature current,
- R_a is armature resistance.

The physical meaning is important. If the motor speeds up, back emf increases. That tends to reduce armature current. If the motor slows under load, back emf falls, armature current rises, and torque increases. This is one reason DC motors are easy to understand in variable-speed applications.

AC motor: main parts

The most common AC motor in practice is the **induction motor**. In homes, this often appears as a single-phase motor in fans, pumps, air coolers, washing machines, and small compressors. In industry, three-phase induction motors are extremely common.

The main parts of a basic induction motor are:

- **stator**,
- **stator winding**,
- **rotor**,

- shaft and bearings,
- frame,
- cooling fan,
- terminal box.

In a **squirrel-cage rotor**, conducting bars are embedded in the rotor and short-circuited by end rings. This construction is rugged and common.

Working principle of an induction motor

In an induction motor, the stator is connected to the AC supply. The stator current produces a magnetic field. In a three-phase motor, this field is naturally rotating. In a single-phase motor, the situation is more subtle and starting assistance is needed in practical designs. The changing or rotating stator field induces current in the rotor conductors. Those rotor currents produce their own magnetic effect, and interaction between stator field and rotor currents produces torque.

This is why the machine is called an **induction motor**: the rotor current is induced, not supplied directly by an external electrical connection in the ordinary cage-rotor case.

Single-phase induction motor idea

Because the local supply in homes and many small shops is 230 V, 50 Hz single-phase AC, single-phase induction motors are common in domestic and light workshop equipment.

A single-phase induction motor usually has:

- a **main winding**,
- an **auxiliary or starting winding**,
- often a **capacitor** in series with the auxiliary winding,

- and a squirrel-cage rotor.

The auxiliary arrangement is used to create a starting condition that gives a rotating-field effect strong enough to start the rotor. Once the motor comes up to speed, the starting winding may be disconnected in some designs, while in capacitor-run designs a capacitor remains in service.

The essential points are:

- single-phase motors are common for domestic and light workshop loads,
- they often need a starting arrangement,
- and they are widely used because single-phase supply is widely available.

Synchronous speed and slip

For AC motors, a central formula is the **synchronous speed** relation:

$$N_s = \frac{120f}{P} \quad (4.9)$$

where:

- N_s is synchronous speed in revolutions per minute (rpm),
- f is supply frequency in hertz,
- P is number of poles.

For example, on a 50 Hz supply:

- a 2-pole machine has $N_s = 3000$ rpm,
- a 4-pole machine has $N_s = 1500$ rpm,
- a 6-pole machine has $N_s = 1000$ rpm.

An induction motor rotor does not normally run exactly at synchronous speed. It runs slightly below it. The difference is de-

scribed by **slip**:

$$s = \frac{N_s - N_r}{N_s} \quad (4.10)$$

If slip is required as a percentage,

$$s\% = \frac{N_s - N_r}{N_s} \times 100 \quad (4.11)$$

Here N_r is the actual rotor speed.

Slip is necessary in an induction motor because rotor current depends on relative motion between the rotating stator field and the rotor. If there were no relative motion at all, induction would vanish and torque would not be maintained in the same way.

Worked Example 4.4

A 4-pole induction motor is connected to a 50 Hz supply. Find its synchronous speed.

Use Equation (4.9):

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4}$$

$$N_s = \frac{6000}{4} = 1500 \text{ rpm}$$

So the synchronous speed is **1500 rpm**.

Worked Example 4.5

A 4-pole, 50 Hz induction motor runs at 1440 rpm. Find the slip percentage.

First find synchronous speed:

$$N_s = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

Now use Equation (4.11):

$$s\% = \frac{1500 - 1440}{1500} \times 100$$

$$s\% = \frac{60}{1500} \times 100 = 4\%$$

So the slip is 4%.

Worked Example 4.6

A DC motor has supply voltage 220 V, back emf 210 V, and armature resistance 1 Ω . Find the armature current.

Use Equation (4.8):

$$V = E_b + I_a R_a$$

So

$$I_a = \frac{V - E_b}{R_a} = \frac{220 - 210}{1} = 10 \text{ A}$$

So the armature current is **10 A**.

This simple result shows why back emf matters. The armature current is not set by supply voltage alone.

AC and DC motors compared

Table 4.1: Introductory comparison of AC and DC motors

Feature	AC motor	DC motor
Supply type	AC	DC
Common field application	Fans, pumps, compressors, industrial drives	Small drives, battery-operated systems, variable-speed drives, traction history
Main special part	Stator winding and induced rotor system	Commutator and brushes in brushed DC motor
Starting idea	Depends on type; induction motor uses stator field and induced rotor current	Torque produced by armature current in magnetic field
Speed concept	Related to frequency and pole number	Related strongly to armature voltage and back emf
Maintenance	Often low for induction motors	Higher in brushed motors because of brushes and commutator

This table is not meant to say one motor is “better” in all cases. The practical choice depends on supply, control need, cost, maintenance, and application.

Nameplate terms

A motor nameplate or catalog commonly includes:

- rated voltage,
- frequency,
- current,
- output power in W or kW,
- speed in rpm,
- phase,
- duty,
- insulation class,
- sometimes capacitor information for single-phase motors.

A transformer label commonly includes:

- primary voltage,
- secondary voltage,
- frequency,
- VA or kVA rating,
- current rating,
- tap information,
- and sometimes enclosure or temperature information.

These basic ratings connect textbook calculations to real equipment selection and safe operation.

Common applications of transformers

Transformers are used in:

- power transmission and distribution,
- battery chargers,
- doorbell circuits,
- welding equipment,

- inverter systems,
- control panels,
- isolation supplies,
- adapter and low-voltage equipment.

Common applications of AC motors

AC motors are used in:

- ceiling fans,
- water pumps,
- air coolers,
- refrigerators and compressors,
- conveyors,
- blowers,
- machine tools,
- workshop grinders and drills,
- industrial process drives.

Common applications of DC motors

DC motors are used in:

- battery-operated equipment,
- toys and small actuators,
- automobile auxiliaries such as wipers and blowers,
- portable tools and small mechanisms,
- some variable-speed drives,
- laboratory demonstration systems.

Worked Interpretation Exercise

Reading a Transformer Specification

Consider the specification table of the Hammond Manufacturing 166N12 power transformer.

The specification lists these important values:

- primary voltage = 115 V, 60 Hz
- secondary voltage = 12.6 V C.T.
- secondary current = 4 A
- power rating = 50.4 VA

Now interpret these one by one.

First, the primary marking 115 V, 60 Hz tells us this transformer is intended for a supply of 115-volt AC at 60 hertz. That means it is not intended to be connected directly to a 230 V, 50 Hz Indian single-phase mains supply. Doing so would over-stress the transformer badly.

Second, 12.6 V C.T. means the secondary is center-tapped. So:

- full secondary voltage from one end to the other is 12.6 V,
- voltage from center tap to either end is approximately 6.3 V.

Third, the current rating 4 A refers to the rated current available from the secondary winding under the specified use conditions.

Fourth, the power rating 50.4 VA matches the secondary voltage and current:

$$VA = V \times I = 12.6 \times 4 = 50.4 \text{ VA}$$

So the listed VA rating is consistent with the voltage and current values on the product page.

This short reading exercise teaches several practical lessons:

- transformer selection begins with input voltage and frequency, not output voltage alone,

- center tap must be interpreted correctly,
- VA rating is checked by voltage-current product,
- and a real catalog entry immediately connects textbook ratios to actual equipment choice.

Further Reading

1. A. E. Fitzgerald, Charles Kingsley Jr., and Stephen D. Umans, *Electric Machinery* — classic reference for transformer principles, DC machines, induction machines, and machine performance.
2. Stephen J. Chapman, *Electric Machinery Fundamentals* — widely used textbook with clear explanations of transformers and AC/DC machines.
3. I. J. Nagrath and D. P. Kothari, *Electric Machines* — popular machine text for transformer calculations, DC machines, and induction motors.
4. P. S. Bimbhra, *Electrical Machinery* — widely used text for transformer and machine fundamentals with many worked examples.

Overview of Basic Semiconductor Devices

{Overview

} This unit introduces semiconductor materials and the fundamental devices built from them, preparing the reader for later study of rectifiers, transistors, switching, and electronic systems.

{Contents

}

Unit 5: Overview of Basic Semiconductor Devices 5.1 Semiconductor Fundamentals 5.2 P-Type and N-Type Semiconductor
5.3 PN Junction Diode 5.4 Transistor

Unit 5: Overview of Basic Semiconductor Devices

Up to this point, the book has focused mainly on electrical quantities, magnetic effects, AC circuits, transformers, and motors. Those topics explain how electrical energy is produced, transferred, measured, and converted into motion. This chapter begins the electronics side of the subject. Here the central question changes. Instead of only asking how current flows in wires and machines, we now ask how a material can be engineered so that current is allowed, opposed, controlled, or amplified inside a tiny solid device.

That is the role of the **semiconductor**. From a semiconductor, engineers build **diodes**, **transistors**, and eventually integrated circuits. The aim here is not semiconductor fabrication science in full detail, but to understand why materials such as silicon are special, how **doping** changes their behavior, how a **p-n junction** forms, why a diode conducts mainly in one direction, and how a transistor can act as a switch or amplifier.

This chapter connects strongly to what has already been learned. Chapter 1 introduced current, voltage, resistance, and waveforms. Chapter 3 showed AC rectification contexts where diodes naturally appear. Chapter 4 ended with practical control and conversion equipment that often contains semiconductor devices. Later chapters on analog circuits, op-amps, and digital logic depend directly

on the ideas introduced here.

5.1 Semiconductor Fundamentals

Why semiconductors are different

Suppose we compare three materials placed in a circuit:

- copper wire,
- glass,
- and silicon.

Copper allows current to pass very easily. Glass practically blocks it. Silicon lies between these two extremes. Its value in electronics comes from the way its conductivity can be changed strongly by temperature, light, electric field, and tiny amounts of impurity atoms called dopants.

This makes semiconductors useful for electronics. A copper wire is excellent for carrying current, but it is not easy to use copper itself as a current-controlling electronic device. A semiconductor can be shaped and doped so that it acts as a rectifier, switch, amplifier, sensor, or logic element.

Conductor, insulator, and semiconductor

A **conductor** is a material in which charge carriers can move easily. Metals such as copper and aluminum are common conductors.

An **insulator** is a material in which charge carriers are not free to move under ordinary conditions. Glass, mica, rubber, and many plastics are insulators.

A **semiconductor** is a material whose conductivity lies between that of a conductor and an insulator.

The three classes are not merely labels of “good,” “bad,” and

“medium” conductors. Their internal energy structure is different, and semiconductor device behavior follows from that structure.

Image prompt for Figure 5.1: Create a clean textbook-style three-panel energy-band diagram comparing a conductor, semiconductor, and insulator. In each panel show a valence band, conduction band, and band gap. For the conductor, show overlap or zero gap between valence and conduction bands. For the semiconductor, show a small band gap. For the insulator, show a large band gap. Label each panel clearly and keep the illustration simple and engineering-oriented.

Energy bands: valence band, conduction band, and band gap

Inside a crystal, electrons do not behave as if each atom were fully isolated. When many atoms come together to form a solid, the discrete energy levels spread into **energy bands**. The highest filled band is called the **valence band**. The next higher band is the **conduction band**. Between them there may be an **energy gap**, also called the **band gap**.

The basic picture is:

- In a **conductor**, electrons can move easily because the valence and conduction situations effectively allow free conduction.
- In a **semiconductor**, the band gap is small enough that some electrons can move to the conduction band under suitable conditions.
- In an **insulator**, the band gap is so large that ordinary conduction is strongly prevented.

Important semiconductors have relatively small band gaps compared with insulators. Silicon, for example, has a band gap of about 1.12 eV. The value itself is less important here than the fact that a

semiconductor band gap is small enough to allow controlled conduction.

Covalent bonding in silicon

Silicon is the most important semiconductor material in basic electronics. A silicon atom has four outer electrons. In a silicon crystal, each atom shares electrons with neighboring atoms, forming **covalent bonds**.

At low temperature, these electrons are largely tied up in bonds, so conduction is weak. As temperature rises, some bonds break. Then some electrons gain enough energy to move into the conduction band. Once an electron leaves a bond, it leaves behind an electron vacancy called a **hole**.

This is a central semiconductor idea:

- the free **electron** can carry negative charge,
- the **hole** behaves like a mobile positive charge carrier.

Both electrons and holes contribute to current in semiconductors.

Intrinsic semiconductor

A very pure semiconductor is called an **intrinsic semiconductor**. In an intrinsic semiconductor, the number of free electrons equals the number of holes.

This equality is physically reasonable. Every time a covalent bond breaks and releases one electron, one hole is left behind. So carriers are generated in pairs:

- one electron,
- one hole.

Pure silicon and pure germanium are standard examples.

Intrinsic semiconductors conduct only weakly at room temperature

compared with metals. That is why pure semiconductor material alone is not enough for most practical electronic devices. The conductivity must be controlled more strongly. This is done by doping.

Extrinsic semiconductor and doping

An impurity-added semiconductor is called an **extrinsic semiconductor**. The deliberate addition of suitable impurity atoms is called **doping**.

Even tiny amounts of suitable impurity atoms can greatly increase conductivity. This shows how sensitive semiconductor behavior is to very small changes in composition.

Doping does not mean “contaminating the material carelessly.” It means introducing carefully chosen impurity atoms in controlled amounts so that charge carriers of the desired type become dominant.

Two important dopant categories are:

- **Donor impurities:** provide extra electrons.
- **Acceptor impurities:** create holes.

These lead to n-type and p-type material, which are studied in the next section.

Why doping matters physically

Without doping, the semiconductor is too weakly conducting for many practical electronic purposes. With doping, the number of useful charge carriers can be increased strongly. More importantly, the engineer can choose whether electrons or holes become the dominant carriers.

This gives a powerful design advantage:

- if we want electron-dominated conduction, we create n-type

material,

- if we want hole-dominated conduction, we create p-type material,
- and if we join p-type and n-type regions together, we create the basis of the diode and transistor.

Temperature effect in semiconductors

An intuition from metals can be misleading in semiconductors. In many metals, heating increases resistance. In semiconductors, increasing temperature can increase the number of available charge carriers, so conductivity tends to increase.

The practical result is:

- in a semiconductor, temperature strongly affects carrier concentration,
- and semiconductor devices must therefore be used within rated temperature limits.

This is one reason datasheets always include temperature ratings.

Worked Example 5.1

A pure semiconductor sample at a certain temperature has created 8×10^{12} free electrons by thermal energy. Assuming it remains intrinsic, how many holes are present?

In an intrinsic semiconductor, electrons and holes are generated in equal numbers.

So:

$$\text{Number of holes} = \text{Number of electrons}$$

Therefore,

$$\text{Number of holes} = 8 \times 10^{12}$$

So the sample contains **8×10^{12} holes**.

This simple question is important because it reinforces the idea that intrinsic carriers are created in electron-hole pairs.

5.2 P-Type and N-Type Semiconductor

From pure silicon to useful material

A pure silicon crystal is important for understanding, but practical devices need more control than intrinsic silicon provides. We now deliberately insert impurity atoms into the crystal. The result is still mostly silicon, but its electrical behavior changes greatly.

The two basic results are:

- **n-type semiconductor**
- **p-type semiconductor**

These are not two unrelated materials. They are two differently doped forms of semiconductor material.

N-type semiconductor

If a pentavalent impurity atom such as phosphorus, arsenic, or antimony is added to silicon, four of its outer electrons form covalent bonds with neighboring silicon atoms. One extra electron remains weakly bound and can become a conduction electron. The dopant acts as a **donor**.

This creates an **n-type semiconductor**, where electrons are the majority carriers.

The name “n-type” does not mean the whole material has net neg-

ative charge. The material remains electrically neutral overall. It means the dominant mobile charge carriers are negative electrons.

P-type semiconductor

If a trivalent impurity atom such as boron, aluminum, or gallium is added to silicon, only three covalent bonds can be completed directly. One bond position lacks an electron. This creates a hole. The impurity acts as an **acceptor**.

This creates a **p-type semiconductor**, where holes are the majority carriers.

Again, “p-type” does not mean the entire material is positively charged. It means the dominant mobile carriers are positive holes.

Majority and minority carriers

This vocabulary is very important in all semiconductor devices.

In n-type material:

- **majority carriers** are electrons,
- **minority carriers** are holes.

In p-type material:

- **majority carriers** are holes,
- **minority carriers** are electrons.

These minority carriers are still important. They become especially important in reverse-biased diodes and transistor operation.

How to picture the carrier situation

The carrier picture is:

- intrinsic semiconductor: balanced but weak conduction,
- n-type semiconductor: many useful electrons added,
- p-type semiconductor: many useful holes created.

So the engineer is not only increasing conductivity. The engineer is choosing which carrier type will dominate.

Image prompt for Figure 5.2: Create a clean textbook-style two-panel illustration comparing n-type and p-type silicon crystals. In the n-type panel, show a silicon lattice with one pentavalent donor atom, four covalent bonds, and one extra free electron. Label donor impurity, free electron, majority carriers, and minority holes. In the p-type panel, show a silicon lattice with one trivalent acceptor atom, one incomplete bond represented as a hole, and label acceptor impurity, hole, majority carriers, and minority electrons.

Properties of n-type and p-type material

The practical properties may be summarized as follows.

Table 5.1: Comparison of n-type and p-type semiconductors

Property	n-type	p-type
Dopant type	pentavalent donor	trivalent acceptor
Typical dopants	P, As, Sb	B, Al, Ga
Majority carriers	electrons	holes
Minority carriers	holes	electrons
Dominant current mechanism	electron conduction	hole conduction

This table is compact, but each row should be connected to the physical carrier picture rather than memorized as isolated words.

Worked Example 5.2

A silicon sample is doped with phosphorus. Identify:

1. whether it becomes p-type or n-type,
2. the majority carriers,
3. the minority carriers.

Phosphorus is a pentavalent impurity. Pentavalent impurities act as donors and create n-type material.

So:

- type = **n-type**
- majority carriers = **electrons**
- minority carriers = **holes**

Practical uses of p-type and n-type material

By themselves, p-type and n-type materials are already useful concepts, but their greatest importance appears when they are combined.

Applications based on p-type and n-type regions include:

- p-n junction diodes,
- bipolar transistors,
- many sensor structures,
- rectifiers,
- switching devices,
- logic and integrated circuits.

P-type and n-type materials are the building blocks from which junction devices are formed.

5.3 PN Junction Diode

From two regions to one junction

Now imagine joining p-type material and n-type material in one crystal. At the instant they are joined, the carrier concentrations are very different on the two sides:

- many holes on the p-side,
- many electrons on the n-side.

Nature does not leave this imbalance unchanged. Carriers begin to diffuse across the boundary. This leads to the most important structure in basic electronics: the **p-n junction**.

Formation of the p-n junction

When p-type and n-type semiconductors are joined, electrons from the n-side and holes from the p-side diffuse toward the junction. Near the boundary, they recombine. As a result, a region forms where there are no mobile charge carriers. This is the **depletion layer**.

This region is electrically very important because it produces a barrier to further majority-carrier diffusion. The junction therefore does not allow free two-way current like an ordinary metal conductor.

Barrier potential and depletion region

The depletion region contains fixed charged ions left behind after mobile carriers diffuse away and recombine. This creates an internal electric field and a built-in potential difference often called the **barrier potential** or **built-in potential**.

The built-in field does not need a mathematical treatment here. Its operating effect is:

- the junction creates a natural barrier,

- forward bias reduces its effective opposition,
- reverse bias increases its effective opposition.

This one idea explains the basic operation of the diode.

Image prompt for Figure 5.3: Create a clean textbook-style illustration of a silicon p-n junction before and after equilibrium. Show p-type region on the left and n-type region on the right. Label holes, electrons, the depletion region, fixed acceptor ions, fixed donor ions, and the built-in electric field direction. Include a simple energy-barrier indication and keep the diagram clear for diploma beginners.

What is a diode

A **diode** is a semiconductor device with two terminals, an **anode** and a **cathode**, that allows current mainly in one direction under ordinary operation.

This one-way behavior is called **rectification**.

Forward bias

If the p-side is connected to the positive terminal of a source and the n-side to the negative terminal, the diode is **forward biased**.

In forward bias:

- the external voltage opposes the barrier,
- the depletion region becomes narrower,
- majority carriers can cross the junction more easily,
- current increases strongly after the forward voltage becomes sufficient.

For a silicon p-n junction diode, conduction begins around a forward voltage of roughly 0.7 V .

This does not mean the diode has exactly 0.700 V under all condi-

tions. The actual forward voltage depends on current, temperature, and device type. But 0.7 V is a useful approximation for an ordinary silicon junction diode.

Reverse bias

If the p-side is connected to the negative terminal and the n-side to the positive terminal, the diode is **reverse biased**.

In reverse bias:

- the external voltage supports the barrier,
- the depletion region widens,
- majority carriers are pulled away from the junction,
- only a very small reverse leakage current flows under normal conditions.

If the reverse voltage becomes too large, breakdown occurs and large current can flow. In ordinary p-n diodes this is usually an unwanted condition unless the diode is specifically designed for breakdown operation, as in a Zener diode.

V-I characteristics of a p-n junction diode

The **V-I characteristic** of a diode tells how current changes with applied voltage.

In forward bias:

- current remains very small at first,
- then rises rapidly after the knee region,
- this is why a diode is often approximated as having a forward drop.

In reverse bias:

- current remains very small over a wide voltage range,
- then rises sharply at breakdown.

For a silicon p-n diode, the forward voltage is often approximated as about 700 mV, while reverse breakdown occurs at a product-dependent voltage that may be tens to hundreds of volts.

Static approximation for basic circuits

For hand calculations, we often use a simplified model:

- diode off in reverse bias,
- diode on in forward bias with about 0.7 V drop for silicon.

This approximation is not the whole truth, but it is very useful for basic circuit analysis.

Worked Example 5.3

A silicon diode is forward biased in a simple circuit. Using the basic constant-drop approximation, what forward voltage may be assumed across the diode?

For an ordinary silicon p-n junction diode, we use:

$$V_D \approx 0.7 \text{ V}$$

So the assumed forward voltage is **about 0.7 V**.

Worked Example 5.4

A silicon diode is connected in series with a resistor and a 5 V DC source. Assume the diode is forward biased and has a constant drop of 0.7 V. Find the voltage across the resistor.

The source voltage is:

$$V_S = 5 \text{ V}$$

The diode drop is:

$$V_D = 0.7 \text{ V}$$

So the resistor voltage is:

$$V_R = V_S - V_D = 5 - 0.7 = 4.3 \text{ V}$$

So the resistor has **4.3 V** across it.

Common diode applications

Several applications are worth knowing.

Rectification: A diode passes one half of an AC waveform more easily than the other. This is the basis of rectifier circuits.

Clipping and limiting: Diodes can limit signal amplitude and protect circuit inputs from excessive voltage.

Reverse battery protection: A series diode can protect a circuit if battery polarity is connected incorrectly.

Signal switching: Fast diodes such as the 1N4148 are used in small-signal switching and logic-related circuits.

Real diode ratings

A real diode datasheet may include:

- maximum reverse voltage,
- continuous forward current,
- forward voltage,
- reverse leakage current,
- package type,
- switching speed or recovery characteristics,
- power dissipation.

These ratings matter because a diode is not “just one-way.” It must

also survive the applied voltage, carry the required current, and operate fast enough for the intended circuit.

5.4 Transistor

Why the transistor matters

The diode allows or blocks current mainly in one direction. The **transistor** goes further. It allows a small input current or voltage condition to control a larger current. This made modern electronics possible.

This section focuses on the **bipolar junction transistor** or BJT, including the NPN and PNP concept and **common-emitter (CE)** operation.

NPN and PNP concept

A BJT is formed from two p-n junctions and has three regions:

- **emitter**
- **base**
- **collector**

There are two types:

- **NPN transistor**
- **PNP transistor**

A bipolar transistor consists of collector, base, and emitter regions, with a very thin base region between emitter and collector.

Image prompt for Figure 5.4: Create a clean textbook-style comparison of NPN and PNP transistor symbols and simplified layer structures. Show the three terminals emitter, base, and collector for each. Label the emitter arrow direction clearly, and place a simple note that arrow out indicates NPN and arrow in indicates

PNP. Include the layer order for each device.

Identification of emitter, base, and collector

Each transistor terminal has a different role.

Emitter: emits the main charge carriers into the base region.

Base: very thin central region that controls the transistor action.

Collector: collects most of the carriers that pass through the base region.

Do not assume collector and emitter are interchangeable. A transistor does not function properly if collector and emitter are reversed, because the internal dopant concentrations are intentionally different.

Pin identification matters.

How a transistor works physically

Consider an NPN transistor. If the base-emitter junction is forward biased and the base-collector junction is reverse biased, carriers injected from the emitter cross the thin base region and are collected by the collector. Because the base is very thin and lightly doped compared with the emitter, most injected carriers reach the collector rather than recombining in the base.

This leads to the central transistor idea:

- a small base current controls a much larger collector current.

In the basic BJT model, a small base current controls a larger collector current of approximately $I_B \times h_{FE}$.

Current relations in a BJT

The basic current relation is:

$$I_E = I_B + I_C \quad (5.1)$$

where:

- I_E is emitter current,
- I_B is base current,
- I_C is collector current.

The DC current gain is often written as:

$$\beta = h_{FE} = \frac{I_C}{I_B} \quad (5.2)$$

So, approximately:

$$I_C = \beta I_B \quad (5.3)$$

This is a simplified relation suitable for basic CE calculations. In real devices, β changes with operating conditions and is never a perfectly fixed constant.

Common-emitter configuration

In the **common-emitter (CE)** configuration, the emitter is the terminal common to the input and output circuits. The input is applied between base and emitter. The output is taken between collector and emitter.

This configuration is widely used because:

- it can provide current gain,
- it can provide voltage gain in amplifier circuits,
- and it is very common in switching applications.

CE mode is best understood in three operating regions.

Cut-off region

In **cut-off**, the base-emitter junction is not sufficiently forward biased. Base current is very small or zero, and collector current is nearly zero. The transistor behaves like an open switch.

Active region

In the **active region**, the base-emitter junction is forward biased and the base-collector junction is reverse biased. Here the transistor acts as an amplifier. Collector current is controlled by base current.

Saturation region

In **saturation**, the transistor is driven hard on. Collector-emitter voltage becomes low, and the transistor behaves like a closed switch.

For switching applications, the practical goal is usually to move the transistor clearly between cut-off and saturation.

CE operation of an NPN transistor

For a typical silicon NPN transistor in CE mode:

- base is about 0.7 V higher than emitter when the transistor is on in the basic approximation,
- collector is usually connected to the supply through a load resistor,
- a small base current allows a larger collector current.

This is why CE circuits are used for:

- LED driving,
- relay driving,
- small-signal amplification,

- sensor interfacing,
- switching logic-level signals to larger currents.

Worked Example 5.5

A transistor in CE mode has current gain $\beta = 100$. If the base current is $20\ \mu\text{A}$, find the collector current using the basic gain relation.

Use Equation (5.3):

$$I_C = \beta I_B$$

Substitute:

$$I_C = 100 \times 20\ \mu\text{A}$$

$$I_C = 2000\ \mu\text{A} = 2\ \text{mA}$$

So the collector current is **2 mA**.

Worked Example 5.6

A transistor in CE mode carries collector current $4\ \text{mA}$ when the base current is $40\ \mu\text{A}$. Find the DC current gain.

Use Equation (5.2):

$$\beta = \frac{I_C}{I_B}$$

Substitute:

$$\beta = \frac{4\ \text{mA}}{40\ \mu\text{A}}$$

Convert 4 mA to 4000 μA :

$$\beta = \frac{4000}{40} = 100$$

So the DC current gain is **100**.

CE transistor as a switch

In a simple switching circuit:

- if base current is absent, transistor is off and collector current is nearly zero,
- if enough base current is provided, transistor saturates and current flows through the collector load.

This makes the transistor an electronically controlled switch.

Practical examples include:

- switching an LED,
- operating a relay from a small control signal,
- driving a buzzer,
- controlling a sensor output stage.

CE transistor as a simple amplifier

In the active region, a small change in base current causes a larger change in collector current. With a collector resistor, this change in current produces a change in output voltage.

This is the basis of amplification:

- small signal at the base,
- larger controlled variation at the collector,
- CE mode can therefore provide useful amplification.

Detailed small-signal transistor amplifier design belongs to a later

electronics course.

Real transistor ratings

A transistor datasheet may include:

- collector-emitter voltage rating,
- collector current rating,
- power dissipation,
- package type,
- DC current gain range,
- saturation voltages,
- pin order.

These ratings tell whether the transistor is suitable for switching or amplification in a given circuit.

Worked Interpretation Exercise

Reading a Transistor Datasheet

Consider the onsemi P2N2222A transistor datasheet.

From the first page and electrical-characteristics table, we can read these values:

- package: T0-92
- terminal order shown on the package drawing: 1 = Collector, 2 = Base, 3 = Emitter
- collector-emitter voltage rating $V_{CE0} = 40 \text{ V}$
- continuous collector current $I_C = 600 \text{ mA}$
- total device dissipation at 25°C ambient: 625 mW
- a typical DC current gain range is listed under various conditions, for example h_{FE} minimum 100 at $I_C = 150 \text{ mA}$, $V_{CE} = 10 \text{ V}$

Now interpret these values carefully.

First, the pin identification matters immediately. The datasheet drawing shows:

- pin 1 = collector
- pin 2 = base
- pin 3 = emitter

This tells us the transistor cannot be wired correctly by guessing the pin order. In a laboratory breadboard circuit, that one detail is often the difference between proper operation and a non-working circuit.

Second, $V_{CE0} = 40\text{ V}$ means the collector-emitter path must not be subjected to more than 40 V under the stated rating condition. So this transistor is suitable for many low-voltage circuits but not for arbitrary high-voltage switching.

Third, $I_C = 600\text{ mA}$ is the continuous collector current rating. That does not mean every circuit should operate right at 600 mA . The actual safe circuit also depends on power dissipation, temperature, and saturation conditions.

Fourth, the current-gain value h_{FE} is not one fixed universal number. It is specified under particular test conditions. This reminds us that transistor gain varies with current and operating point. Rough calculations may use a nominal value, but practical design always respects datasheet ranges.

Fifth, saturation values such as $V_{CE(sat)}$ directly support the switching idea from this chapter: when driven hard on, the transistor does not become a perfect short circuit, but the collector-emitter voltage becomes low.

This one artifact ties several chapter ideas together:

- identification of collector, base, and emitter,
- current gain,
- switching use,
- voltage and current limits,
- and the difference between a conceptual transistor model and a real component.

Further Reading

1. Robert L. Boylestad and Louis Nashelsky, *Electronic Devices and Circuit Theory* — popular text for diode and transistor fundamentals with practical circuit examples.
2. Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits* — standard electronics text covering semiconductor devices, diode circuits, and BJT operation.
3. Ben G. Streetman and Sanjay Kumar Banerjee, *Solid State Electronic Devices* — widely used semiconductor-device text for energy bands, doping, p-n junctions, and device physics.
4. S. M. Sze and Kwok K. Ng, *Physics of Semiconductor Devices* — classic reference for deeper semiconductor-device physics and p-n junction behavior.

Overview of Analog Circuits

{Overview

} This unit surveys foundational analog-circuit ideas such as amplification, biasing, signal handling, and the role of analog building blocks in instrumentation and control.

{Contents

}

Unit 6: Overview of Analog Circuits 6.1 Op-Amp Fundamentals
6.2 Basic Op-Amp Configurations 6.3 Op-Amp Applications

Unit 6: Overview of Analog Circuits

The previous chapter introduced semiconductor devices such as diodes and transistors. Those devices showed how solid-state electronics can control current. This chapter takes the next step and introduces one of the most important analog building blocks in electronics: the **operational amplifier**, usually called the **op-amp**. An op-amp is a very high-gain differential amplifier that becomes useful when we apply **negative feedback** around it. With only a few resistors, the same op-amp can be made to invert a signal, amplify it without inversion, add several signals together, buffer a weak sensor source, or shift a low-level signal into a more useful range. This flexibility is the reason op-amps appear throughout analog electronics, instrumentation, signal conditioning, control circuits, and interface stages.

This chapter stays within the syllabus boundary. We focus on the ideal op-amp idea, the representative 741C pin configuration, the concept of **virtual ground**, the inverting and non-inverting amplifier, gain calculation, the op-amp adder, and a few typical low-level signal-conditioning uses. The treatment is application-oriented.

6.1 Op-Amp Fundamentals

Why an op-amp is useful

Suppose a sensor gives a signal of only 50 mV, but the next stage of the system works much better if it receives about 1 V. Or suppose one part of a circuit must measure a signal without loading it heavily. Or suppose two low-level analog signals need to be added together before measurement. A transistor can help in some cases, but circuit design becomes easier if we use a ready-made amplifier block whose behavior is predictable.

That block is the op-amp.

An op-amp is an active circuit element designed with characteristics such as very high input resistance, low output resistance, and large differential gain, which make it a very useful building block in many analog circuits.

The op-amp as a differential amplifier

An op-amp has two input terminals:

- **non-inverting input**, usually marked +
- **inverting input**, usually marked –

and one output terminal.

It responds to the **difference** between the two input voltages. If we call the non-inverting input voltage V_+ and the inverting input voltage V_- , then the open-loop output idea is

$$V_o = A_{OL}(V_+ - V_-) \quad (6.1)$$

where A_{OL} is the **open-loop gain**.

This one equation explains why the op-amp is so powerful and also why it is usually not used in open loop for ordinary linear amplification. If A_{OL} is very large, even a tiny difference between the

input terminals can drive the output strongly.

Open-loop and closed-loop idea

If an op-amp is used **open loop**, there is no feedback path from output to input. Since the open-loop gain is very large, the output usually saturates near one supply limit or the other even for a very small input difference.

If the op-amp is used with **negative feedback**, part of the output is fed back in such a way that an increase in output tends to reduce the effective input difference. This makes the circuit stable and predictable for linear analog work.

This distinction drives the chapter:

- open loop gives huge gain but poor linear usefulness,
- closed loop with negative feedback gives controlled, usable gain.

Image prompt for Figure 6.1: Create a clean textbook-style diagram comparing an open-loop op-amp and a negative-feedback op-amp. In the left panel, show an op-amp with inputs V_+ and V_- and no feedback, with the output saturating high or low for a tiny input difference. In the right panel, show an op-amp with a feedback resistor path from output to the inverting input and indicate stable closed-loop operation. Label non-inverting input, inverting input, output, and negative feedback clearly.

Features of an ideal op-amp

The ideal op-amp is a teaching model. For basic op-amp gain analysis, the usual assumptions are:

1. infinite open-loop gain,
2. infinite input impedance,

3. zero output impedance.

These assumptions are extremely useful because they simplify circuit analysis. Many textbooks also add two more ideal ideas:

- infinite bandwidth,
- zero input current.

We can summarize the most useful ideal properties this way:

Table 6.1: Ideal op-amp assumptions

Ideal property	Meaning in simple words
Infinite open-loop gain	Even a very tiny input difference can produce a large output response
Infinite input impedance	The op-amp inputs draw no current ideally
Zero output impedance	The output behaves like an ideal voltage source
Infinite bandwidth	Gain does not fall with frequency in the ideal model
Infinite CMRR and infinite PSRR	Common-mode signals and supply changes do not affect the ideal output

The first three assumptions are enough for most gain derivations in this chapter.

What high input impedance means physically

If the input impedance is very high, the source connected to the input does not need to supply much current into the op-amp input.

This means the op-amp does not heavily **load** the source. With very little loading on the source, the op-amp can act as a buffer. This is especially important in signal conditioning. Many sensors do not like to be loaded by a low-resistance circuit.

What low output impedance means physically

If the output impedance is low, the op-amp can drive the next stage more easily without the output voltage dropping badly because of internal resistance. This is one reason a voltage follower or buffer is useful even though its gain may be only 1.

Real op-amp versus ideal op-amp

Real op-amps are not ideal. They have:

- finite gain,
- finite bandwidth,
- finite slew rate,
- nonzero offset voltage,
- nonzero input bias current,
- limited output swing,
- and limited supply range.

This chapter still uses the ideal model for most calculations because it helps you understand the circuit structure first. Then the representative 741C is used to show how a real op-amp is specified.

The 741C as a representative op-amp IC

The syllabus specifically asks for the 741C pin configuration. The 741C should be treated as a representative historical and educational op-amp, not as the only modern choice. The LM741 is a single, 44-V, 1-MHz operational amplifier, and newer alternatives are

available with better characteristics.

So the 741C remains useful for learning:

- symbol and pin identification,
- dual-supply operation idea,
- general-purpose op-amp behavior,
- and classic analog-circuit examples.

But in modern practical design, many newer op-amps are preferred for lower power, wider input range, rail-to-rail operation, lower offset, or single-supply use.

741C pin configuration

The LM741 8-pin package uses the following pin functions:

Table 6.2: LM741C representative pin functions

Pin number	Function
1	Offset null
2	Inverting input
3	Non-inverting input
4	Negative supply V_-
5	Offset null
6	Output
7	Positive supply V_+
8	No connection

The pinout shows that the op-amp needs supply pins in addition to signal pins. The triangle symbol alone is not enough; the physical device must be powered.

Image prompt for Figure 6.2: Create a clean textbook-style top-view pin diagram of the LM741C in an 8-pin DIP package. Label pin 1 offset null, pin 2 inverting input, pin 3 non-inverting input, pin 4 negative supply, pin 5 offset null, pin 6 output, pin 7 positive supply, and pin 8 no connection. Include the notch orientation clearly.

Typical 741C characteristics

Typical LM741 values include:

- gain-bandwidth product is typically 1 MHz,
- slew rate is typically 0.5 V/ μ s,
- input offset voltage is a few millivolts,
- typical supply current is about 1.7 mA,
- LM741C recommended operating temperature range is 0°C to 70°C,
- recommended supply is typically dual supply around ± 15 V, though the device can operate over a range.

A real op-amp is defined by many limits beyond just “gain,” so specification reading must include speed, offset, supply range, and output limits.

Power supply idea for op-amps

The classical 741 is usually shown with a positive and a negative supply, such as +15 V and -15 V. This allows the output to swing both positive and negative around ground.

Modern op-amps may also work on a single supply, such as 0 V and +5 V. That is very common in sensor and microcontroller circuits. The general op-amp concept remains the same, but biasing and allowable input/output ranges must then be handled carefully.

Virtual short and virtual ground

This is one of the most useful concepts in op-amp analysis.

A **virtual short** is the condition in which the non-inverting and inverting inputs have almost the same voltage when a high-gain op-amp is used with negative feedback.

So under ideal negative-feedback operation:

$$V_+ \approx V_- \quad (6.2)$$

This does **not** mean the two inputs are physically connected together. It means the input difference becomes extremely small because the op-amp's large open-loop gain and feedback force the circuit to that condition.

If one of the inputs is actually connected to ground, and the other input becomes almost equal to it because of the virtual short condition, that other node is called a **virtual ground**.

So for an inverting amplifier with the non-inverting input grounded:

$$V_+ = 0 \quad \Rightarrow \quad V_- \approx 0$$

but the inverting node is still not physically tied to ground.

This concept makes gain derivation much easier.

Virtual open idea

Because the ideal op-amp has infinite input impedance, no current enters either input terminal:

$$I_+ = I_- = 0 \quad (6.3)$$

Students often use the phrase **virtual open** for this input-current

idea. Together, the ideal op-amp rules most often used in circuit analysis are:

- input currents are zero,
- input voltages are nearly equal under negative feedback.

Those two rules will now be used repeatedly.

Worked Example 6.1

An ideal op-amp with negative feedback has its non-inverting input connected to ground. If the circuit is operating linearly, what voltage may be assumed at the inverting input for ideal analysis?

Under negative feedback, ideal op-amp analysis uses the virtual short:

$$V_- \approx V_+$$

Since

$$V_+ = 0$$

therefore

$$V_- \approx 0$$

So the inverting input is treated as being at **virtual ground**, approximately 0 V.

6.2 Basic Op-Amp Configurations

The most striking thing about op-amps is that many useful functions come from only a few external resistors. In this section we study the two most important basic forms:

- the **inverting amplifier**
- the **non-inverting amplifier**

If these are understood clearly, later applications such as adders, filters, level shifters, and sensor interfaces become much easier to understand.

Inverting amplifier

In an **inverting amplifier**, the signal is applied through an input resistor to the inverting input. The non-inverting input is usually connected to ground or to a reference level. A feedback resistor connects the output back to the inverting input.

An inverting amplifier reverses the input signal polarity and applies a defined gain set by the resistor network.

Image prompt for Figure 6.3: Create a clean textbook-style schematic of an inverting op-amp amplifier using an ideal op-amp. Show input voltage V_i connected through resistor R_1 to the inverting input, feedback resistor R_f from output to the inverting input, the non-inverting input grounded, output labeled V_o , and dual supplies indicated separately. Label the inverting input node as virtual ground.

Let:

- input resistor = R_1
- feedback resistor = R_f
- input voltage = V_i

- output voltage = V_o

Because of virtual ground,

$$V_- \approx 0$$

Since no current enters the op-amp input, the current through R_1 equals the current through R_f .

Current through input resistor:

$$I_1 = \frac{V_i - 0}{R_1} = \frac{V_i}{R_1}$$

Current through feedback resistor:

$$I_f = \frac{0 - V_o}{R_f} = -\frac{V_o}{R_f}$$

Since $I_1 = I_f$,

$$\frac{V_i}{R_1} = -\frac{V_o}{R_f}$$

Therefore,

$$\boxed{\frac{V_o}{V_i} = -\frac{R_f}{R_1}} \quad (6.4)$$

So the **voltage gain** of the inverting amplifier is

$$\boxed{A_v = -\frac{R_f}{R_1}} \quad (6.5)$$

The negative sign means the output is 180° out of phase with the input. A positive input change produces a negative output change, and vice versa.

Physical meaning of the inverting amplifier

The inverting amplifier does two things at once:

- it changes the signal magnitude according to the resistor ratio,
- it reverses the polarity or phase of the signal.

This circuit is very common when signal inversion is acceptable or desired, and when multiple input signals must later be summed into one node, as in the adder circuit.

Input impedance of the inverting amplifier

The input signal of an inverting amplifier typically comes from a low-impedance source because the input impedance of this circuit is determined mainly by the input resistor.

The source connected to the inverting amplifier “sees” approximately:

$$Z_{in} \approx R_1$$

That is different from the non-inverting amplifier, which has much higher input resistance.

Worked Example 6.2

An inverting amplifier has $R_1 = 10 \text{ k}\Omega$ and $R_f = 50 \text{ k}\Omega$. Find the voltage gain.

Use Equation (6.5):

$$A_v = -\frac{R_f}{R_1} = -\frac{50 \text{ k}\Omega}{10 \text{ k}\Omega}$$

$$A_v = -5$$

So the voltage gain is **-5 V/V**.

This means the output magnitude is five times the input magnitude, but with inversion.

Worked Example 6.3

An inverting amplifier has gain -4 . If the input signal is 0.3 V, find the output voltage.

Use:

$$V_o = A_v V_i$$

So:

$$V_o = -4 \times 0.3 = -1.2 \text{ V}$$

So the output voltage is **-1.2 V**.

Non-inverting amplifier

In a **non-inverting amplifier**, the input signal is applied directly to the non-inverting input. The inverting input is connected to a feedback divider formed by two resistors.

The non-inverting amplifier has the desirable property of high input resistance, making it useful for buffering nonideal sources while also providing gain greater than one.

Image prompt for Figure 6.4: Create a clean textbook-style schematic of a non-inverting op-amp amplifier using an ideal op-amp. Show input voltage V_i connected to the non-inverting input, resistor R_1 from inverting input to ground, feedback resistor R_f from output to the inverting input, and output labeled V_o .

Indicate the resistor divider at the inverting input clearly.

For the non-inverting amplifier, the inverting input voltage is the divided version of the output:

$$V_- = V_o \cdot \frac{R_1}{R_1 + R_f}$$

Using the virtual short under negative feedback:

$$V_- \approx V_+ = V_i$$

So:

$$V_i = V_o \cdot \frac{R_1}{R_1 + R_f}$$

Rearranging,

$$V_o = V_i \cdot \frac{R_1 + R_f}{R_1}$$

Therefore,

$$\boxed{\frac{V_o}{V_i} = 1 + \frac{R_f}{R_1}} \quad (6.6)$$

and the voltage gain is

$$\boxed{A_v = 1 + \frac{R_f}{R_1}} \quad (6.7)$$

Notice that the gain is positive. So the output stays in phase with the input.

Physical meaning of the non-inverting amplifier

This circuit is especially useful when:

- the source should not be loaded much,
- the output should keep the same polarity as the input,
- and a moderate or large gain is required.

Because the input signal is connected directly to the op-amp non-inverting input, the source benefits from the op-amp's high input resistance.

Worked Example 6.4

A non-inverting amplifier has $R_1 = 5 \text{ k}\Omega$ and $R_f = 45 \text{ k}\Omega$.

Find the voltage gain.

Using Equation (6.7):

$$A_v = 1 + \frac{R_f}{R_1} = 1 + \frac{45}{5}$$

$$A_v = 1 + 9 = 10$$

So the voltage gain is **+10 V/V**.

Worked Example 6.5

A non-inverting amplifier has gain +6. If the input signal is 0.2 V , find the output voltage.

Use:

$$V_o = A_v V_i$$

So:

$$V_o = 6 \times 0.2 = 1.2 \text{ V}$$

So the output voltage is **+1.2 V**.

Table 6.3: Comparison of basic amplifier configurations

Feature	Inverting amplifier	Non-inverting amplifier
Input terminal used	inverting input via resistor	non-inverting input directly
Output phase	inverted	same phase
Gain formula	$-R_f/R_1$	$1 + R_f/R_1$
Input resistance seen by source	approximately R_1	very high
Good use	scaling with inversion, summing	buffering and scaling without inversion

This table is worth understanding deeply because many practical analog circuits are built from these two patterns.

Virtual ground revisited in the inverting amplifier

Students often see the inverting-input node labeled 0 V and mistakenly think it is directly grounded. It is not. It is at approximately zero volts because of negative feedback and the op-amp’s large gain. If the circuit saturates or the feedback conditions fail, the virtual-ground assumption also fails.

So virtual ground is a **working condition**, not a physical connection.

Output saturation

Real op-amps cannot produce arbitrarily large output voltage. The output is limited by the supply rails and internal voltage drops. In many cases, the actual output limits are smaller than the supply voltages because of internal drops.

So even if the gain formula predicts a certain output, the real circuit may clip or saturate if the required output exceeds the allowed range.

6.3 Op-Amp Applications

Why applications grow naturally from the basic forms

Many op-amp applications are variations of the circuits already studied. The adder, subtractor, active filter, comparator interface, buffer, and many sensor-conditioning circuits all grow from the same negative-feedback ideas.

The syllabus asks specifically for:

- **adder using op-amp**
- **typical low-level signal conditioning applications**

The following sections treat those applications in a practical way.

Summing amplifier or adder

If several input voltages are connected to the inverting input through separate resistors, while feedback is provided from output to that same node, the op-amp can produce an output proportional to the sum of the input currents. This is called a **summing amplifier** or **adder**.

A summing amplifier is a basic inverting amplifier with multiple

inputs, where the output is a linear sum of the input contributions, each weighted by its resistor ratio.

Image prompt for Figure 6.5: Create a clean textbook-style schematic of an inverting summing amplifier. Show three input voltages V_1 , V_2 , and V_3 applied through resistors R_1 , R_2 , and R_3 to the inverting input of an ideal op-amp. Show feedback resistor R_f from output to the inverting input. Show the non-inverting input grounded. Label the inverting node as virtual ground and the output as V_o .

Using virtual ground, the current through each input resistor is:

$$I_1 = \frac{V_1}{R_1}, \quad I_2 = \frac{V_2}{R_2}, \quad I_3 = \frac{V_3}{R_3}$$

Since no current enters the op-amp input, these currents add and flow through the feedback resistor:

$$I_f = I_1 + I_2 + I_3$$

The feedback current is also:

$$I_f = -\frac{V_o}{R_f}$$

Therefore,

$$-\frac{V_o}{R_f} = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}$$

So the general output is:

$$\boxed{V_o = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)} \quad (6.8)$$

If all input resistors are equal, say

$$R_1 = R_2 = R_3 = R$$

then

$$V_o = -\frac{R_f}{R}(V_1 + V_2 + V_3) \quad (6.9)$$

If also $R_f = R$, then the circuit becomes a simple inverting adder:

$$V_o = -(V_1 + V_2 + V_3) \quad (6.10)$$

Worked Example 6.6

An op-amp adder has three equal input resistors of $10 \text{ k}\Omega$ and a feedback resistor of $10 \text{ k}\Omega$. The input voltages are:

- $V_1 = 0.2 \text{ V}$
- $V_2 = 0.5 \text{ V}$
- $V_3 = 0.1 \text{ V}$

Find the output voltage.

Since all input resistors are equal and $R_f = R$, use Equation (6.10):

$$V_o = -(V_1 + V_2 + V_3)$$

So:

$$V_o = -(0.2 + 0.5 + 0.1) = -0.8 \text{ V}$$

So the output voltage is **-0.8 V**.

Weighted adder idea

If the input resistors are not equal, then each signal contributes a different weight to the output. That is useful in:

- mixing signals,
- combining sensor channels,
- creating analog computing functions,
- and digital-to-analog style resistor-weighted circuits.

Resistor values set how strongly each input affects the output, which is why the summing amplifier is useful for mixing and weighting signals.

Typical low-level signal conditioning

Signal conditioning means preparing a signal so that the next stage can use it properly. In practice, op-amps are often used to:

- buffer a weak sensor output,
- amplify a low-level signal,
- shift a signal into a desired range,
- filter unwanted noise together with suitable RC networks,
- combine more than one signal.

Op-amp circuits often translate low-level sensor outputs into ranges suitable for ADC inputs, using inverting or non-inverting amplifier forms with reference levels.

The core low-level signal-conditioning uses are buffering and scaling.

Buffering a weak source

A **buffer** or **voltage follower** has gain approximately equal to 1:

$$V_o \approx V_i \quad (6.11)$$

This may look useless at first because it does not increase signal amplitude. But because of high input resistance and low output resistance, it isolates the source from the load.

A sensor that cannot supply much current may be able to feed a buffer successfully. The next stage then receives the same voltage but from a stronger output source.

Scaling a small sensor signal

Suppose a temperature sensor, pressure sensor, photodiode conditioner, or bridge output stage gives only a few millivolts or a few hundred millivolts. The signal may be too small for the next measuring or processing stage. An op-amp amplifier can scale it upward.

For example, an op-amp can translate a sensor output to a more usable range for an ADC.

The physical idea is:

- the sensor provides the information,
- the op-amp reshapes the electrical level into a more useful form.

Worked Example 6.7

A low-level sensor gives an output of 50 mV. The next stage works best with about 1.0 V. Design a simple non-inverting gain value that would scale 50 mV to 1.0 V.

Required gain:

$$A_v = \frac{V_o}{V_i} = \frac{1.0}{0.05} = 20$$

For a non-inverting amplifier:

$$A_v = 1 + \frac{R_f}{R_1}$$

So:

$$20 = 1 + \frac{R_f}{R_1}$$

$$\frac{R_f}{R_1} = 19$$

One simple resistor choice is:

- $R_1 = 1 \text{ k}\Omega$
- $R_f = 19 \text{ k}\Omega$

So a **non-inverting gain of 20** will scale 50 mV to 1.0 V.

Why the 741C is only illustrative

The syllabus correctly notes that the 741C is an illustrative example, not the only modern device. The 741 has several practical limitations:

- is not rail-to-rail,
- often prefers dual supplies,
- has modest speed,
- and has characteristics that are ordinary compared with many modern op-amps.

Many newer alternatives offer lower quiescent current and better performance.

So the correct learning attitude is:

- understand the 741C well,
- but understand the **general op-amp concept** even more deeply.

Worked Interpretation Exercise

Reading an Op-Amp Datasheet

Consider the TI LM741 datasheet and product information.

From the pin-configuration and specification tables, we can read:

- pin 2 = inverting input
- pin 3 = non-inverting input
- pin 4 = negative supply
- pin 6 = output
- pin 7 = positive supply
- recommended LM741C operating temperature range = 0°C to 70°C
- typical gain-bandwidth product = 1 MHz
- typical slew rate = $0.5\text{ V}/\mu\text{s}$
- typical supply current = about 1.7 mA

Now interpret these values in a practical way.

First, the pin table tells us the 741 cannot be wired just by knowing the triangle symbol. The physical IC package must be connected correctly. If the supply pins are forgotten, the amplifier will not operate at all.

Second, the presence of both positive and negative supply pins shows the classical dual-supply style of many textbook op-amp circuits. This helps explain why classic examples often use +15 V, -15 V, and ground.

Third, the 1 MHz gain-bandwidth value tells us the 741 is not a high-speed modern op-amp. It is suitable for learning and for many low-frequency analog tasks, but not for every modern fast-signal application.

Fourth, the $0.5 \text{ V}/\mu\text{s}$ slew rate reminds us that the output cannot change infinitely fast. Even if the ideal model suggests an instantaneous response, the real device has a finite rate of change.

Fifth, the 0°C to 70°C range associated with the 741C version explains the meaning of the “C” grade in basic educational use. This is one reason the chapter treats the 741C as a representative device, not a universal solution for all environments.

This short interpretation exercise connects textbook ideas with a real op-amp component:

- symbol versus package pinout,
- ideal model versus real specifications,
- and why datasheet reading matters before circuit use.

Further Reading

1. Adel S. Sedra and Kenneth C. Smith, *Microelectronic Circuits* — standard text for op-amp models, negative feedback, amplifier configurations, and analog applications.
2. Paul Horowitz and Winfield Hill, *The Art of Electronics* — influential practical electronics text with strong treatment of op-amps and analog circuit design.
3. Sergio Franco, *Design with Operational Amplifiers and Analog Integrated Circuits* — dedicated op-amp text covering inverting, non-inverting, summing, and signal-conditioning circuits.
4. Robert F. Coughlin and Frederick F. Driscoll, *Operational Amplifiers and Linear Integrated Circuits* — popular text for 741-style op-amp circuits and linear IC applications.

Overview of Digital Electronics

{Overview

} This unit introduces digital-electronics fundamentals, including logic levels, gates, Boolean reasoning, combinational building blocks, and the basic interpretation of digital systems.

{Contents

}

Unit 7: Overview of Digital Electronics 7.1 Logic Gates 7.2
Boolean Algebra 7.3 Standard Theorems and Forms

Unit 7: Overview of Digital Electronics

The earlier chapters dealt mainly with quantities that can vary continuously. Voltage may rise smoothly, current may fall gradually, and an op-amp may amplify a small analog signal into a larger analog signal. Digital electronics works differently. Instead of treating every intermediate value as meaningful, it groups signals into a small number of logical states, usually two: 0 and 1.

This simple idea makes it possible to build reliable systems that can count, compare, decide, store, and control. A temperature controller, a washing-machine sequence circuit, a battery charger indicator, a traffic-light controller, and the logic section of a microcontroller-based system all depend on digital decisions. Digital electronics is built from **logic gates**, **Boolean algebra**, and standard expression forms such as **SOP** and **POS**.

This chapter explains those foundations carefully. We begin with the physical idea of digital signals and logic levels. We then study the common gates and their truth tables, learn how Boolean algebra simplifies logic expressions, apply **De Morgan's theorem**, and build expressions in **sum-of-products** and **product-of-sums** form. These foundations prepare the later study of combinational circuits, sequential circuits, counters, and programmable logic.

7.1 Logic Gates

Suppose a switch-controlled lamp circuit is designed so that the lamp is treated as **OFF** when the control voltage is near 0 V and **ON** when the control voltage is near 5 V. In a real circuit, the voltage may not be exactly 0 V or exactly 5 V at every instant. There can be noise, wiring drop, or switching delay. Even so, digital design works well because the circuit does not need infinitely precise values. It only needs the voltage to fall within a range recognized as logic LOW or logic HIGH.

This is the key abstraction of digital electronics. A varying electrical signal is interpreted as one of a few logical states.

A **digital signal** is a signal that represents information using discrete levels. In basic digital electronics, we usually use two levels:

- logic 0, also called **LOW**
- logic 1, also called **HIGH**

The actual voltage corresponding to LOW and HIGH depends on the logic family and supply voltage. For example, the TI SN74HC00 NAND gate operates from 2 V to 6 V, and its recommended input thresholds depend on the supply voltage. At $V_{CC} = 4.5$ V, its input thresholds are $V_{IL(max)} = 1.35$ V and $V_{IH(min)} = 3.15$ V. That means an input not above 1.35 V is safely LOW, and an input not below 3.15 V is safely HIGH.

So digital electronics is not the claim that only two voltages exist. It is the design method of assigning ranges of voltage to two logical meanings.

Positive logic

In **positive logic**, the higher voltage level represents logic 1 and the lower voltage level represents logic 0. This chapter uses positive logic unless stated otherwise. Logic-gate datasheets commonly state the Boolean function “in positive logic” when describing a device.

For example, if a gate output is near the positive supply, we interpret that as logic 1. If the output is near ground, we interpret that as logic 0.

Why digital circuits are robust

The advantage of this method is practical robustness. If a signal that should be HIGH is 4.7 V one moment and 4.9 V the next, the logic meaning remains the same. Small unwanted variations are ignored as long as the signal stays within the valid region.

This is one reason digital systems became so important in control and computation. The circuit makes decisions based on state, not on tiny analog differences.

Truth tables

A **truth table** lists all possible input combinations of a logic function and shows the corresponding output. It is one of the most useful tools in digital electronics because it gives the complete behavior of a gate or logic expression in an organized way.

For a single-input gate, there are 2 possible input states: 0 and 1.

For a two-input gate, there are $2^2 = 4$ input combinations:

- 00
- 01
- 10

- 11

For a three-input gate, there are $2^3 = 8$ combinations.

If a logic function has n binary inputs, the truth table contains 2^n rows.

A switching picture of logic

Before naming the common gates, it helps to connect them to ordinary switching ideas.

- An **AND** action is like two series switches. The load receives power only if both switches are closed.
- An **OR** action is like two parallel switches. The load receives power if either switch is closed.
- A **NOT** action is an inversion. If the input command is 1, the output becomes 0; if the input command is 0, the output becomes 1.

This does not mean that a digital IC literally contains mechanical switches. It means the logical behavior can first be understood with a familiar physical analogy.

Image prompt for Figure 7.1: Create a clean textbook-style illustration with three panels. Panel 1 shows an AND action using two series switches controlling a lamp. Panel 2 shows an OR action using two parallel switches controlling a lamp. Panel 3 shows a NOT action using a control block labeled inverter that produces the opposite output state. Label input switches A and B, lamp state, and logic outputs clearly.

AND gate

An **AND gate** gives a HIGH output only when all its inputs are HIGH.

For a two-input AND gate:

$$Y = A \cdot B \tag{7.1}$$

Here A and B are the input variables and Y is the output. The dot means logical multiplication, also called the AND operation.

Table 7.1: Truth table of a two-input AND gate

A	B	Y = A·B
0	0	0
0	1	0
1	0	0
1	1	1

An everyday interpretation is a motor starter interlock. Suppose a small control system should energize a relay only when the START signal is HIGH and the SAFETY_OK signal is HIGH. That is an AND condition.

OR gate

An **OR gate** gives a HIGH output when at least one input is HIGH. For a two-input OR gate:

$$Y = A + B \tag{7.2}$$

The plus sign here does not mean ordinary arithmetic addition. It means logical OR.

Table 7.2: Truth table of a two-input OR gate

A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1

If either of two sensors is allowed to trigger an alarm, the output logic can be OR.

NOT gate

A **NOT gate** has one input and one output. It produces the complement of the input.

$$Y = \bar{A} \quad (7.3)$$

The bar over a variable means **complement** or **logical inverse**.

Table 7.3: Truth table of a NOT gate

A	Y = $\bar{A}$
0	1
1	0

If a signal named `DOOR_OPEN` is 1 when the door is open, then $\bar{A}$ represents the logic meaning “door not open,” which in a simple two-state system means “door closed.”

NAND gate

A **NAND gate** is a NOT-AND combination. It gives the complement of the AND output.

$$Y = \overline{A \cdot B} \tag{7.4}$$

So a NAND gate is LOW only when both inputs are HIGH. In every other case, it is HIGH.

Table 7.4: Truth table of a two-input NAND gate

A	B	Y = (A·B)
0	0	1
0	1	1
1	0	1
1	1	0

Because it combines logic action with inversion, NAND becomes extremely useful in practical digital systems.

NOR gate

A **NOR gate** is a NOT-OR combination. It gives the complement of the OR output.

$$Y = \overline{A + B} \tag{7.5}$$

So a NOR gate is HIGH only when all inputs are LOW.

Table 7.5: Truth table of a two-input NOR gate

A	B	$Y = (A+B)$
0	0	1
0	1	0
1	0	0
1	1	0

EX-OR gate

The **exclusive-OR** gate, written **EX-OR** or **XOR**, gives a HIGH output when the inputs are different.

$$Y = A \oplus B = \bar{A}B + A\bar{B} \quad (7.6)$$

This equation says that the output is HIGH in either of two cases:

- $A = 0, B = 1$
- $A = 1, B = 0$

Table 7.6: Truth table of a two-input EX-OR gate

A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

An XOR gate is often described as an **inequality detector** for two inputs. If the two inputs differ, the output is 1.

EX-NOR gate

The **exclusive-NOR** gate, written **EX-NOR** or **XNOR**, gives a HIGH output when the inputs are equal.

$$Y = \overline{A \oplus B} = AB + \bar{A}\bar{B} \tag{7.7}$$

Table 7.7: Truth table of a two-input EX-NOR gate

A	B	Y = (A ⊕ B)
0	0	1
0	1	0
1	0	0
1	1	1

So XNOR is an **equality detector**. When two digital signals should match, XNOR is a natural comparison function.

A combined comparison table

Table 7.8: Summary of common two-input gates

Gate	Boolean expression	Output is HIGH when...	Simple meaning
AND	$A \cdot B$	both inputs are HIGH	all conditions satisfied
OR	$A+B$	at least one input is HIGH	any condition satisfied
NAND	$(A \cdot B)^{-}$	all cases except both HIGH	NOT-AND
NOR	$(A+B)^{-}$	both inputs are LOW	NOT-OR
XOR	$A \oplus B$	inputs are different	inequality detector
XNOR	$(A \oplus B)^{-}$	inputs are same	equality detector

Worked Example 7.1

A safety interlock produces output $Y = A \cdot B$, where:

- $A = 1$ means the protective door is closed,
- $B = 1$ means the emergency stop is released.

Find the output for the following conditions.

1. $A = 1, B = 1$
2. $A = 1, B = 0$
3. $A = 0, B = 1$

Using the AND rule:

$$Y = A \cdot B$$

Case 1:

$$Y = 1 \cdot 1 = 1$$

Case 2:

$$Y = 1 \cdot 0 = 0$$

Case 3:

$$Y = 0 \cdot 1 = 0$$

So the machine is enabled only in Case 1, when both conditions are satisfied.

Worked Example 7.2

A comparison circuit uses $Y = A \oplus B$. Find the output for all four input combinations and explain the meaning.

From the XOR truth table:

- $0 \oplus 0 = 0$
- $0 \oplus 1 = 1$
- $1 \oplus 0 = 1$
- $1 \oplus 1 = 0$

So the output is HIGH only when the inputs are unequal.

Universal logic gates

The NAND and NOR gates are called **universal gates** because any basic logic function can be built using only NAND gates or only NOR gates.

This is not just a mathematical curiosity. It has practical importance. If a designer has a stock of one gate type, many logic functions can still be realized.

Realizing NOT with NAND

If both inputs of a NAND gate are tied together and fed with the same variable A, then

$$Y = \overline{A \cdot A} = \bar{A} \quad (7.8)$$

So a NAND gate can work as an inverter.

Realizing AND with NAND

First generate $\overline{A \cdot B}$ using one NAND gate, then invert it with a second NAND gate used as a NOT gate:

$$Y = \overline{\overline{A \cdot B}} = A \cdot B \quad (7.9)$$

Realizing OR with NAND

Using De Morgan's theorem,

$$A + B = \overline{\bar{A} \cdot \bar{B}} \quad (7.10)$$

This means:

1. invert A using a NAND-as-NOT,
2. invert B using another NAND-as-NOT,
3. NAND the two inverted signals.

Then the final output becomes $A + B$.

Realizing NOT with NOR

If both inputs of a NOR gate are tied together,

$$Y = \overline{A + A} = \bar{A} \quad (7.11)$$

Realizing OR with NOR

First form $\overline{A+B}$ using one NOR gate. Then invert it using a second NOR-as-NOT stage:

$$Y = \overline{\overline{A + B}} = A + B \quad (7.12)$$

Realizing AND with NOR

Using De Morgan's theorem,

$$A \cdot B = \overline{\bar{A} + \bar{B}} \quad (7.13)$$

So by first inverting A and B with NOR gates and then NORing those inverted outputs, we obtain an AND function.

Image prompt for Figure 7.2: Create a clean textbook-style logic illustration showing universal-gate realization. In one column, show

NOT, AND, and OR implemented using only NAND gates. In a second column, show NOT, OR, and AND implemented using only NOR gates. Label each intermediate node and final Boolean function clearly.

Floating inputs must be avoided

Leaving an unused CMOS input open is unsafe. Slow or floating CMOS inputs can cause extra current and unpredictable behavior. In practical circuits, unused logic inputs are tied to a defined HIGH or LOW level according to the application.

In laboratory work, if a breadboarded logic circuit behaves erratically, one of the first things to check is whether every input has a valid logic level.

7.2 Boolean Algebra

Why Boolean algebra is needed

Truth tables show complete behavior, but they become longer as the number of inputs increases. A circuit with four inputs already requires $2^4 = 16$ rows. For larger systems, we need a symbolic way to express logic and simplify it. That symbolic language is **Boolean algebra**.

Boolean algebra deals with variables that take only two values:

- 0
- 1

Unlike ordinary algebra, Boolean algebra uses logical operations such as AND, OR, and NOT.

Boolean variables and literals

A **Boolean variable** is a variable that can have only the values 0 or 1.

Examples:

- A
- B
- X
- Y

A variable itself, such as A, is called a **literal**. Its complement, such as $\bar{A}$, is also a literal.

Basic postulates and laws

The most useful Boolean laws are listed in one place below.

Table 7.9: Basic Boolean laws

Law name	Boolean form	Meaning
Identity for OR	$A + 0 = A$	OR with 0 changes nothing
Identity for AND	$A \cdot 1 = A$	AND with 1 changes nothing
Null law for OR	$A + 1 = 1$	OR with 1 always gives 1
Null law for AND	$A \cdot 0 = 0$	AND with 0 always gives 0
Idempotent OR	$A + A = A$	repeating the same term does not change result

Law name	Boolean form	Meaning
Idempotent AND	$A \cdot A = A$	same
Complement OR	$A + \bar{A} = 1$	one of them must be true
Complement AND	$A \cdot \bar{A} = 0$	both cannot be true together
Involution	$\bar{\bar{A}} = A$	double inversion returns original value
Commutative OR	$A + B = B + A$	order does not matter
Commutative AND	$A \cdot B = B \cdot A$	order does not matter
Associative OR	$(A+B)+C = A+(B+C)$	grouping does not matter
Associative AND	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$	grouping does not matter
Distributive	$A(B+C) = AB+AC$	AND distributes over OR
Distributive	$A+BC = (A+B)(A+C)$	OR distributes over AND

These laws are the tools used in simplification.

Physical meaning of some basic laws

The law $A + 1 = 1$ says that if one OR input is already HIGH, the output is certainly HIGH, whatever the other input may be.

The law $A \cdot 0 = 0$ says that if one AND input is LOW, the output is certainly LOW, whatever the other input may be.

The law $A + \bar{A} = 1$ says that either a statement or its complement must be true in a two-state system.

These are simple, but they save a lot of work in actual logic reduction.

Simple algebraic simplification

Consider the expression

$$Y = A + A \cdot B$$

We can simplify it using a standard absorption result:

$$A + A \cdot B = A \tag{7.14}$$

Why is this correct? If $A = 1$, then the output is certainly 1. If $A = 0$, then $A \cdot B = 0$, so the whole expression is still 0. Therefore B does not affect the result.

Another important simplification is

$$A(A + B) = A \tag{7.15}$$

Again, once A is known, the extra term does not change the final result.

Worked Example 7.3

Simplify

$$Y = A + A\bar{B}$$

Using the absorption law:

$$A + A\bar{B} = A$$

So the simplified expression is

$$Y = A$$

Worked Example 7.4

Simplify

$$Y = A \cdot 1 + B \cdot 0$$

Using the identity and null laws:

$$A \cdot 1 = A$$

and

$$B \cdot 0 = 0$$

Therefore

$$Y = A + 0 = A$$

Worked Example 7.5

Simplify

$$Y = A\bar{A} + B$$

Since

$$A\bar{A} = 0$$

we get

$$Y = 0 + B = B$$

Truth-table verification of an algebraic result

In Boolean algebra, a simplification can always be checked by a truth table. This habit prevents symbol mistakes.

For example, verify

$$A + A\bar{B} = A$$

Table 7.10: Verification of $A + A\bar{B} = A$

A	B	$\bar{B}$	$A\bar{B}$	$A + A\bar{B}$	A
0	0	1	0	0	0
0	1	0	0	0	0
1	0	1	1	1	1
1	1	0	0	1	1

The two final columns are identical. So the simplification is correct.

7.3 Standard Theorems and Forms

De Morgan's theorem

One of the most useful theorems in digital electronics is **De Morgan's theorem**. It provides a systematic way to handle complements of expressions.

For two variables, the theorem has two forms:

$$\overline{A \cdot B} = \bar{A} + \bar{B} \quad (7.16)$$

and

$$\overline{A + B} = \bar{A} \cdot \bar{B} \quad (7.17)$$

The rule in words is simple:

- the complement of a product becomes the sum of the complements,
- the complement of a sum becomes the product of the complements.

De Morgan's theorem is essential when converting between NAND/NOR implementations and standard Boolean expressions.

Why De Morgan's theorem matters physically

Suppose an output should go LOW only when both conditions are true. That is a NAND function. Instead of thinking only in terms of “NOT after AND,” De Morgan's theorem lets us rewrite the same function as “the first condition is false OR the second condition is false.” Both descriptions are logically equivalent.

This theorem often allows a circuit to be built more economically from available gate types.

Worked Example 7.6

Apply De Morgan's theorem to simplify

$$Y = \overline{A + B}$$

Using Equation (7.17),

$$Y = \bar{A}\bar{B}$$

So a complemented OR expression is equal to the AND of the complemented variables.

Worked Example 7.7

Rewrite

$$Y = \overline{AB}$$

using De Morgan's theorem.

From Equation (7.16),

$$Y = \bar{A} + \bar{B}$$

Truth-table verification of De Morgan's theorem

Table 7.11: Verification of $\overline{A+B} = \bar{A}\bar{B}$

A	B	A+B	$\overline{(A+B)}$	$\bar{A}$	$\bar{B}$	$\bar{A}\bar{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0

A	B	$A+B$	$(A+B)'$	$\bar{A}$	$\bar{B}$	$\bar{A}\cdot\bar{B}$
1	1	1	0	0	0	0

The columns for $(A+B)'$ and $\bar{A}\cdot\bar{B}$ are identical, so the theorem is verified.

Minterms

A **minterm** is a standard product term in which every variable appears once, either in direct form or complemented form.

For two variables A and B, the four minterms are:

- $\bar{A}\bar{B}$
- $\bar{A}B$
- $A\bar{B}$
- AB

Each minterm is equal to 1 for exactly one row of the truth table.

Table 7.12: Two-variable minterms

A	B	Minterm that is 1
0	0	$\bar{A}\bar{B}$
0	1	$\bar{A}B$
1	0	$A\bar{B}$
1	1	AB

This is why minterms are useful. They allow us to build a Boolean expression directly from the rows where the output is 1.

SOP representation

SOP means **sum of products**. Each product term is usually a minterm or a simplified product term, and the final expression is formed by ORing those product terms.

To write the canonical SOP of a logic function:

1. write the rows of the truth table for which the output is 1,
2. write the corresponding minterm for each such row,
3. OR the minterms together.

Worked Example 7.8

A function Y of variables A and B has output 1 for the rows:

- $A = 0, B = 1$
- $A = 1, B = 0$

Write the SOP expression.

For 01, the minterm is

$$\bar{A}B$$

For 10, the minterm is

$$A\bar{B}$$

So the SOP expression is

$$Y = \bar{A}B + A\bar{B} \quad (7.18)$$

This is exactly the XOR function.

Worked Example 7.9

The truth table of a function $Y(A, B, C)$ gives output 1 for the rows:

- 001
- 011
- 110

Write the canonical SOP.

For 001, the minterm is

$$\bar{A}\bar{B}C$$

For 011, the minterm is

$$\bar{A}BC$$

For 110, the minterm is

$$AB\bar{C}$$

Therefore

$$Y = \bar{A}\bar{B}C + \bar{A}BC + AB\bar{C} \quad (7.19)$$

This is the canonical SOP form.

Maxterms

A **maxterm** is a standard sum term in which every variable appears once, either in direct or complemented form.

Each maxterm is equal to 0 for exactly one row of the truth table.

For two variables A and B, the four maxterms are:

- $A + B$

- $A + \bar{B}$
- $\bar{A} + B$
- $\bar{A} + \bar{B}$

The pattern is the opposite of minterms. In a maxterm, a variable appears uncomplemented when its row value is 0, and complemented when its row value is 1, so that the whole sum becomes 0 only for that row.

Table 7.13: Two-variable maxterms

A	B	Maxterm that is 0
0	0	$A + B$
0	1	$A + \bar{B}$
1	0	$\bar{A} + B$
1	1	$\bar{A} + \bar{B}$

POS representation

POS means **product of sums**. The canonical POS is formed from the rows for which the output is 0.

To write the canonical POS:

1. identify the rows where the output is 0,
2. write the corresponding maxterm for each such row,
3. AND those sum terms together.

Worked Example 7.10

Suppose a two-variable function Y is 0 for the rows:

- 00
- 11

Write the canonical POS.

For 00, the maxterm is

$$A + B$$

For 11, the maxterm is

$$\bar{A} + \bar{B}$$

Therefore

$$Y = (A + B)(\bar{A} + \bar{B}) \quad (7.20)$$

Worked Example 7.11

A function $Y(A, B, C)$ is 0 for the rows:

- 000
- 010
- 111

Write the canonical POS.

For 000, the maxterm is

$$A + B + C$$

For 010, the maxterm is

$$A + \bar{B} + C$$

For 111, the maxterm is

$$\bar{A} + \bar{B} + \bar{C}$$

So the canonical POS is

$$Y = (A + B + C)(A + \bar{B} + C)(\bar{A} + \bar{B} + \bar{C}) \quad (7.21)$$

Relation between SOP and POS

SOP is built from the rows where the output is 1.

POS is built from the rows where the output is 0.

Both describe the same logic function, but from opposite viewpoints. In practice:

- SOP often matches an OR of AND-gate outputs,
- POS often matches an AND of OR-gate outputs.

This makes the forms especially useful when connecting algebra to gate-level realization.

Simple conversion and verification example

Consider the function with truth table:

Table 7.14: Example truth table for conversion

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

From the rows where $Y = 1$, the SOP is

$$Y = \bar{A}B + A\bar{B}$$

From the rows where $Y = 0$, the POS is

$$Y = (A + B)(\bar{A} + \bar{B})$$

These look different, but they describe the same function. A truth table or algebraic expansion can verify the equivalence.

Worked Example 7.12

Verify that

$$\bar{A}B + A\bar{B}$$

and

$$(A + B)(\bar{A} + \bar{B})$$

represent the same logic function.

Expand the POS form:

$$(A + B)(\bar{A} + \bar{B}) = A\bar{A} + A\bar{B} + B\bar{A} + B\bar{B}$$

Now use

$$A\bar{A} = 0, \quad B\bar{B} = 0$$

So

$$(A + B)(\bar{A} + \bar{B}) = A\bar{B} + \bar{A}B$$

which is exactly the XOR SOP expression.

Therefore the two forms are equivalent.

Worked Interpretation Exercise

Reading a Real Logic-IC Datasheet: TI SN74HC00

The TI SN74HC00 is a useful real artifact for this chapter because it is a standard logic-gate IC containing four independent 2-input NAND gates.

The most useful information is:

1. Part description

The device contains **four independent 2-input NAND gates** and performs the Boolean function in positive logic. That means one IC package gives four identical NAND-gate sections.

So if a circuit needs up to four NAND gates, one SN74HC00 may be enough.

2. Supply-voltage range

The recommended operating supply range is 2 V to 6 V. This tells us immediately that the device is from the HC CMOS logic family, not a 230 V mains device and not a raw transistor that can simply be connected without a supply plan.

A regulated 5 V supply is a common operating choice.

3. Package and pin count

The PDIP version has 14 pins. The datasheet shows the functional pinout. Important pins include:

- 1A, 1B as the inputs of gate 1
- 1Y as the output of gate 1
- 2A, 2B, 2Y for gate 2
- 3A, 3B, 3Y for gate 3
- 4A, 4B, 4Y for gate 4

- GND on pin 7
- VCC on pin 14

This tells us how to wire the IC on a breadboard. Without checking VCC and GND, the logic function on paper cannot become a working circuit.

4. Input thresholds

At $V_{CC} = 4.5 \text{ V}$, the input thresholds are:

- $V_{IL(max)} = 1.35 \text{ V}$
- $V_{IH(min)} = 3.15 \text{ V}$

So an input up to 1.35 V is safely read as LOW, and an input from 3.15 V upward is safely read as HIGH. Voltages in between are not reliable logic levels.

This is a practical warning. It is tempting to assume that 2.0 V should “probably be HIGH enough” in a 5 V logic system, but the threshold values show why that assumption is unsafe.

5. Fanout and family behavior

The device supports fanout up to 10 LSTTL loads. One gate output can drive several compatible logic inputs, but output loading is still a real design consideration.

Further Reading

1. M. Morris Mano and Michael D. Ciletti, *Digital Design* — standard text for Boolean algebra, truth tables, logic simplification, and combinational/sequential design.
2. Thomas L. Floyd, *Digital Fundamentals* — popular introductory reference for logic gates, Boolean laws, standard forms,

and laboratory-oriented digital electronics.

3. John F. Wakerly, *Digital Design: Principles and Practices* — widely used text connecting logic theory, gates, timing, and digital system design.
4. Claude E. Shannon, *A Symbolic Analysis of Relay and Switching Circuits* — landmark paper connecting Boolean algebra with switching circuits.